

# Standingless Interiors and Residual Coherence

A Kernel-Faithful Account of Governance Deficiency in Horizon-Isolated Carriers

AASC Black-Hole Endpoint Arc, Paper II: Standingless Interior Bridge

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## Abstract

This paper is an AASC-native structural extrapolation that presupposes the AASC Einstein projection rather than rejecting it. It does not compute black-hole interior dynamics, replace general relativity, predict infalling experience, identify a quantum-gravity microdynamics, or claim that the interior is empty. Its target is narrower: the role-status of crossed content in an uncertified permanent-horizon case once local metric validity is distinguished from standing transport.

The central control result is that a black-hole horizon is not an AMetric reset. The AMetric boundary is an admissibility-generating boundary upstream of metricity; a black-hole horizon, considered as a same-scope horizon inside the AASC-GR projection, is downstream of metric comparison, causal typing, exterior/interior role structure, and standing-governed reconstruction. Retyping such a horizon as a fundamental licensing surface is possible only as declared new governance, certificate, richer scope, hidden import, repair, or bookkeeping. It is not ordinary same-scope horizon crossing.

Given the AASC kernel, standing bivalence, horizon-standing break, no horizon-as-AMetric-reset, and the absence of leakage, non-sink quotient, certificate, declared high-curvature/effective load, or richer governance, the horizon-isolated carrier cannot instantiate a same-scope standing-bearing physical interior for crossed content. Local AASC-GR metric continuation does not decide this standing question: a licensed metric equation may remain the correct local projection while failing to supply standing transport, standing recovery, or AMetric re-licensing.

The paper introduces role-indexed residual coherence, written  $\text{ResCoh}_D^r(x)$ , as the constrained non-standing persistence of formerly governed role-organization. It requires pre-break role ancestry, post-break standing loss, no new governance or certificate, non-null representation, retained organization, and failure of full role licensing. Standing is bivalent; residual coherence is role-indexed and gradable.

Because measurement standing depends on metric comparison, scale anchoring, calibration continuity, witness and record fixation, reference stability, and re-fixability or admissible witness-equivalence, measurement-tied roles are residualized whenever those supports lack standing. Local notation, tensor writability, and source-ledger form do not by themselves supply standing-bearing measurement inside the horizon-isolated carrier. Where measurement-like organization remains representable, its status is residual metrological coherence rather than full same-scope measurement standing.

The resulting interior is neither an unqualified continuation of ordinary metric-matter interiority nor simple nullity. It is a governance-deficient residual-coherence carrier. Skin and representative roles have weakest residual support. Tensor-like and source/metric roles, where representable, retain stronger residual organization without standing. Metricity, matter-energy, time, and causality

remain compatible with AASC-GR as local projection roles, but for crossed content lacking standing transport they no longer retain full same-scope role licensing. Governance-depth names residual successor-support thinning, not radial distance, proper time, curvature magnitude, or a dynamical decay process.

## AASC Black-Hole Endpoint Arc Position

*Shared arc spine: kernel/AMetric invariant  $\rightarrow$  licensed AASC-GR black-hole regime  $\rightarrow$  exterior no-totalization  $\rightarrow$  standingless residual interior  $\rightarrow$  governance-depth exhaustion  $\rightarrow B_A^* \rightarrow$  AMetric homology  $\rightarrow$  fresh internal governed domain.*

**Shared series abstract.** The AASC Black-Hole Endpoint Arc is a five-manuscript sequence that treats black holes as a fixed-domain stress test for AASC. It first role-types the licensed black-hole regime, then proves the exterior no-totalization obstruction, then classifies the uncertified horizon-isolated carrier as residual coherence rather than ordinary standing-bearing interiority, then exposes the endpoint AMetric normal form and no-crossing homology, and finally classifies the counter-boundary survivor as fresh internal governed-domain status. The sequence is conditional and one-way: later endpoint or capstone conclusions never authorize earlier target formation.

**Paper label. Paper II / Standingless interior bridge.** This manuscript classifies the uncertified horizon-isolated carrier after standing break as residual coherence rather than ordinary same-scope interiority.

This manuscript is Paper II of the arc. It is the interior-side bridge between exterior no-totalization and endpoint boundary exposure. Its job is to push the standing break inward without confusing local metric continuation with standing-preserving governance.

**Imported into this paper.** The paper imports the exterior no-totalization result: crossed standing is not transported through an uncertified permanent horizon, and  $E_{\text{int}}$  is the horizon-isolated carrier on which old-scope standing transport is undefined. It also imports the AASC-GR projection as a licensed local metric projection, not as a standing certificate.

**Proved here.** The paper proves that, absent leakage, certificate, non-sink quotient, declared high-curvature/effective load, or new governance scope,  $E_{\text{int}}$  is not an ordinary same-scope standing-bearing physical interior for crossed content. The horizon is not an AMetric reset. What remains, where representable, is role-indexed residual coherence: skin, tensor-like load, metricity, measurement, matter-energy, time, and causality residualize according to pre-break role load and governance-depth.

**Exported to the next paper.** The export is residual-coherence status, the no-horizon-as-AMetric-reset lock, the metric-continuation/standing-continuation separation, and governance-depth as residual successor-support thinning. These are the inputs needed for endpoint-side role-support exhaustion.

**Not claimed here.** This paper does not claim local GR failure, local device malfunction, matter annihilation, interior nullity, or a dynamical decay process. It also does not yet prove that the endpoint is  $B_A^*$ , does not prove AMetric normal-form homology, and does not infer fresh internal governed-domain status.

**Series dependency lock.** The only black-hole arc input used here is the exterior standing break: crossed content lacks old-scope standing transport on  $E_{\text{int}}$ . This paper adds the no-horizon-as-AMetric-reset lock and the residual-coherence classification. It does not use endpoint AMetric homology, no-crossing, or fresh-domain recurrence as premises.

**Forward handoff.** The endpoint paper may use this export only in the following controlled form:

$$\begin{aligned} \text{Stand}_D(x) &= 0, \\ \text{ResCoh}_D^r(x) &\text{ where role-indexed organization remains representable,} \\ \text{Succ}_D^{\text{res}}(x, \text{metric}) &= \emptyset \text{ only at residual metric-support exhaustion.} \end{aligned}$$

Thus residual coherence is not itself  $B_A^*$ . It becomes endpoint-relevant only when the relevant residual successor support is exhausted under the fixed target.

**Certificate terminology lock.** The word *certificate* has four arc-level uses, and the uses are not interchangeable. A *standing-transport certificate* is a same-scope map or witness assignment that preserves a threatened reference class in the exterior reconstruction problem. An *exterior witness certificate* is a concrete exterior, boundary, radiation, remnant, holographic, island, or quotient map that removes a burden from the unresolved-residue ledger. An *endpoint-certified target* is not a rescue certificate; it is the antecedent package in which a nonempty unresolved residue, old-scope standing loss, residual metric-support exhaustion, closed false endpoints, null-route closure, and no prior re-governance make endpoint AMetric exposure the target. A *capstone-certified endpoint* is the Paper IV input: the homology handoff together with endpoint certification and no-crossing. A horizon can be transport-certified and thereby exit the exterior obstruction; an unresolved residue can be endpoint-certified precisely because no such transport certificate has resolved it.

**Exterior-quotient and metric-regime lock.** The proof-level object of the arc is the declared exterior standing quotient of the black-hole model, not a particular metric-regime representative. In the familiar settled Kerr–Newman example the quotient is displayed by mass, electric charge, and angular momentum, but that example does not restrict the theorem to stationary representatives and does not make absence of hair a separate proof premise. Hair-like, dynamical, evaporating, ringdown, perturbative, semiclassical, quasi-local, high-curvature, or quantum-gravity structure has same-scope force only when it is declared as exterior quotient content, an exterior witness, a leakage channel, a certificate, a non-sink quotient, or richer governance. Such structure subtracts from the unresolved residue exactly to the extent that it re-fixes standing-bearing content. If it does not re-fix standing, it is skin, bookkeeping, hidden load, repair, selector import, carrier laundering, or scope change; it does not provide an additional endpoint route.

**Metric-regime and hair residualization lock.** The standingless-interior theorem does not depend on stationarity as a proof-level condition and does not deny hair by assumption. Stationarity, ringdown, perturbation, evaporation, trapping-horizon structure, local smoothness, high-curvature development, and semiclassical dynamics are metric or envelope descriptors inside an already licensed projection. Hair-like structure is treated by the same role rule. If it is a standing-bearing exterior witness or certificate, it re-governs the content it actually re-fixes and subtracts that content before the residual-coherence stage. If it remains a hidden or representative feature without

standing-bearing exterior effect, it is residual organization, skin, hidden load, bookkeeping, repair, selector import, or scope change. It does not create same-scope standing inside  $E_{\text{int}}$  merely by being describable.

**Referee-facing arc firewall.** The residual-coherence theorem is not a disguised claim of ignorance, local catastrophe, or hidden nullity. It applies only after the exterior paper has supplied the old-scope standing break and only while no certificate, leakage route, non-sink quotient, declared high-curvature/effective load, or new governance scope has been supplied. A richer theory may avoid the theorem by re-governing the carrier or by mapping it back to the old standing class; in that case the result is a certificate or scope change, not a refutation of the residual classification.

The firewall also separates three notions that hostile readings often collapse. Local metric continuation is not standing continuation. Residual coherence is not partial standing. Governance-depth is not radial depth, proper time, affine parameter, curvature magnitude, or a physical decay process. This paper therefore does not prove  $B_A^*$ , AMetric homology, no-crossing, or fresh-domain status. It exports only the residual-support vocabulary and the condition under which residual metric-comparison support may later exhaust.

**Pass 6 notation lock.** The carrier  $E_{\text{int}}$  is imported from Paper I only as the old-scope horizon-isolated carrier for crossed content after standing transport fails. Residual coherence is written  $\text{ResCoh}_D^r(x)$ , where  $r$  is a role index such as skin, tensor-like load, metric, measurement, source, time, or causal dependence.  $\text{ResCoh}_D^r(x)$  is not standing  $\Sigma$ , not entropy, not information, and not a third standing value. Residual successor support is  $\text{Succ}_D^{\text{res}}(x, r)$ , and governance-depth  $\text{GDepth}_D(x, r)$  is defined by thinning of that support rather than by radius, proper time, affine parameter, or curvature. This paper exports residual metric-support exhaustion as a possible handoff condition; it does not itself introduce the endpoint token  $B_A^*$ .

**Pass 9 local referee-response lock.** For local review, residual coherence should not be read as ignorance, approximate ordinary interior physics, nullity, or partial standing. It is a constrained status requiring pre-break role ancestry, post-break standing loss, no certificate or new governance, non-null representability, retained organization, and failure of full role licensing. The matter-energy and measurement claims are likewise role-license claims: they do not assert local annihilation, empirical malfunction, or failure of the AASC–GR projection. This paper exports only residual-coherence status and the governance-depth vocabulary. It does not expose the New AMetric Boundary; that later step requires the additional condition that residual metric-comparison successor support is empty.

**Pass 10 hostile-referee closure lock.** The hostile objection that residual coherence is merely ignorance, hidden ontology, or partial standing is closed by its six-part definition: role ancestry, standing loss, no certificate or new governance, non-null representation, retained organization, and failure of full role licensing. The objection that the result contradicts GR is closed by the AASC–GR compatibility lock: local metric projection may remain valid while standing transport fails. The objection that an infalling observer or local apparatus restores ordinary interior measurement is closed unless a calibration bridge, witness ledger, or new governance scope is supplied. Thus a real counterexample must either show that physical roles are primitive rather than governance-dependent,



or supply declared re-governance/certification; otherwise the residual-coherence classification remains the old-scope survivor.

## Main theorem and claim discipline

The central theorem is the following:

**Standingless Interior and Residual Coherence Theorem.** Given the AASC kernel, bivalent standing, no same-scope standing transport or standing creation across an uncertified permanent horizon, the fact that a black-hole horizon is a derivative metric boundary rather than an AMetric licensing boundary, a horizon-isolated carrier  $E_{\text{int}}$ , no leakage, certificate, non-sink quotient, or declared new governance scope, and the dependence of realized physical interior roles on governance, crossed content does not enter  $E_{\text{int}}$  as a same-scope standing-bearing physical interior. Any retained describable role organization, where representable, has role-indexed residual-coherence status rather than standing-bearing role status. Skin has weakest residual persistence; tensor-like load, where representable, retains stronger residual organization without standing; for crossed content lacking standing transport, and where role-indexed organization remains representable, metric, measurement, matter, energy, time, and causality no longer retain full same-scope role licensing; measurement-tied quantities persist, where representable, only as residual metrological coherence; and what metric projection calls depth is, at the structural level, governance-depth until residual metric comparison itself loses support.

The claim is strong but conditional. It is not a theorem of classical general relativity alone and not a dynamical quantum-gravity model. It is a kernel-level AASC result about what may be claimed of a horizon-isolated carrier once standing has been broken and no admissible re-governance has been supplied.

| Claim                                                             | Status in this paper                       |
|-------------------------------------------------------------------|--------------------------------------------|
| Standing is bivalent                                              | Kernel commitment                          |
| Standing gradually weakens                                        | Denied                                     |
| Crossed standing is transported into $E_{\text{int}}$             | Denied absent certificate                  |
| The interior creates standing to replace the loss                 | Denied absent new governance               |
| The black-hole horizon functions as an AMetric reset              | Denied                                     |
| The horizon licenses standing in the same scope                   | Denied; it is a derivative metric boundary |
| $E_{\text{int}}$ is absolute nothingness                          | Not claimed                                |
| $E_{\text{int}}$ remains an ordinary same-scope physical interior | Denied absent certificate/new governance   |
| Non-standing physical load may remain describable                 | Claimed as residual coherence              |
| Residual coherence is role-indexed and gradable                   | Main construction                          |
| Skin has weakest residual persistence                             | Main theorem                               |
| Tensor-like load persists longer as residual organization         | Main theorem                               |

| Claim                                                                               | Status in this paper                                     |
|-------------------------------------------------------------------------------------|----------------------------------------------------------|
| Metricity remains fully licensed without governance                                 | Denied                                                   |
| Local notation or tensor writability is measurement standing                        | Denied                                                   |
| Measurement-tied roles remain fully licensed without metric/comparison standing     | Denied absent certificate or new metrological governance |
| Measurement-like organization may remain representable                              | Claimed as residual metrological coherence               |
| Matter and energy are primitive and survive unchanged                               | Denied                                                   |
| Matter-energy roles, where representable, have residual/governance-deficient status | Main theorem                                             |
| Physical time and causality remain primitive                                        | Denied                                                   |
| Metric depth remains stable at all governance-depth                                 | Denied                                                   |
| The paper predicts infalling experience                                             | Not claimed                                              |
| The paper solves black-hole interiors dynamically                                   | Not claimed                                              |

## Reader's guide

This manuscript is a companion to the exterior black-hole reconstruction theorem. That theorem is intentionally cautious: it proves that an uncertified permanent horizon cannot be a continuing same-scope exterior reconstruction object when threatened exterior standing lacks leakage, certificate, or non-sink quotient. The present paper pushes the interior consequence as far as the kernel permits. It asks: if standing is not transported into the horizon-isolated carrier and is not created there, can the carrier still be an ordinary standing-bearing physical interior in the old same-scope sense?

The answer is no. A new interior governance structure may be declared; a holographic or boundary map may re-couple the carrier; a fuzzball-like replacement, island-like reconstruction domain, remnant map, or other quantum-gravity mechanism may supply different governance. Those are not rejected. But if none is declared, the carrier cannot inherit ordinary physical interiority from the standing that failed to cross. It is not a hidden ordinary spacetime region full of fully licensed matter and energy. It is a residual-coherence regime.

A central control point is that the black-hole horizon is not an AMetric reset. Standing is initially licensed through governance admission downstream of the AMetric boundary, but that boundary is prior to metricity. A black-hole horizon is downstream of metricity and already presupposes the governance structure it would need to generate. Treating it as a new licensing boundary is therefore a new scope or hidden import, not same-scope horizon crossing.

For non-AASC readers, the main terms can be read as follows. Standing means licensed eligibility for later identity-preserving use. Re-fixability means that later data can identify the same reference

class under declared redescription rules. Governance means the package of admissibility, standing, reference, and irreversibility required for non-degenerate continuation. Skin is presentation without independent load. Tensor-like load is structure that had load-bearing content before standing broke. Residual coherence is what remains of such role-organization after standing is gone. Measurement standing means more than a local interaction or writable quantity: it requires comparison, scale, calibration, witness/record fixation, reference, and re-fixability or admissible witness-equivalence in the declared scope.

## Status of the argument

This paper is an AASC-native extrapolation theorem. It is not an independent derivation of black-hole interior dynamics from general relativity, semiclassical gravity, or quantum gravity. Its claims are conditional on the AASC kernel, the AMetric-to-metric licensing order, the horizon obstruction result, and the realized physical interior dependency stack. The contribution of the paper is to classify what the horizon-isolated carrier can be, under those premises, once the horizon is excluded as an AMetric reset.

The argument is therefore conditional but not merely verbal. The classifications below are forced by dependency order. A black-hole horizon is derivative because its identity depends on metric and causal governance. Residual coherence is constrained by pre-break role ancestry, post-break non-null representation, retained organization, standing loss, absence of certificate or new governance, and failure of full role licensing. New interior governance requires a declared kernel tuple. These conditions prevent residual coherence from becoming a name for whatever remains and prevent the horizon from being treated as an undeclared source of standing.

| Dependency                      | Function in this paper                                                |
|---------------------------------|-----------------------------------------------------------------------|
| AASC kernel $K_D$               | supplies admissibility, standing, reference, and irreversibility      |
| AMetric boundary                | supplies initial governance licensing; it is upstream of metricity    |
| Horizon obstruction             | supplies standing-transport failure in the uncertified permanent case |
| No-reset theorem                | excludes the same-scope horizon as a new licensing boundary           |
| Realized physical interior      | shows ordinary physical roles require governance support              |
| Metric projection               | shows metricity is downstream of comparison and standing              |
| Mass/inertia/source ledger      | supports matter-energy residualization as role-license failure        |
| Arrow/time/causality discipline | supports temporal and causal residualization                          |
| Information/record boundary     | blocks information-storage as hidden standing recovery                |

## Comparator orientation

Standard references on classical horizons, quasi-local horizon frameworks, black-hole thermodynamics, holographic reconstruction, quantum extremal surfaces, and island formulas are used here as comparator orientation rather than proof authority. They help locate the target against event horizons, trapping and isolated horizons, holographic screens, quantum extremal surfaces, island reconstruction, and semiclassical information-recovery proposals. The AASC theorem itself is carried by the kernel, AASC-GR projection, no-reset result, horizon obstruction, and residual-coherence classification.

## Part I - Boundary and Kernel Discipline

### After the Standing Break

The exterior reconstruction theorem isolates a horizon obstruction. In a fixed black-hole exterior reconstruction scope, a threatened reference class may become dependent on a candidate permanent horizon. If no leakage, radiation witness, boundary trace, holographic anchor map, complementarity correspondence, remnant map, non-sink quotient, or other certificate is supplied, then the reference class has no admissible future image in the old exterior witness algebra. The horizon standing-transport operator is partial, and any same-scope totalization is repair, selector import, carrier laundering, bookkeeping, or scope change.

The present paper begins after that result. It does not ask whether a metric diagram can display an interior. It asks whether the carrier that appears as interior can instantiate the same kind of non-degenerate physical interior that the realized-physics architecture requires. The answer depends on whether standing remains available. If standing does not cross and is not created, then the old physical role architecture cannot simply continue as though nothing governance-level has happened.

The strongest kernel-faithful interpretation is not that the interior is empty. It is also not that the interior is ordinary but inaccessible. It is that, absent re-governance, the horizon-isolated carrier is a residual-coherence carrier. It contains whatever non-null load remains describable after standing has been broken, but that load is no longer licensed as ordinary standing-bearing metric-matter content in the same-scope AASC sense.

### The target question

The target question is:

In a horizon-isolated carrier where crossed standing is neither transported nor internally created, what becomes of the metric, matter, energy, mass, temporal, and causal roles that normally require standing-governance?

This question is not answered by saying that general relativity has an interior coordinate region. Metricity is not primitive. It is a licensed comparison role. It is also not answered by saying that the interior contains matter or energy. Matter and energy are not primitive. They are field, particle, gauge, deformation-response, transport, source-ledger, and metric-facing roles. If those roles require governance, then a standingless carrier cannot be treated as their ordinary continuation without explanation.

## The central answer

The answer is residual coherence. Crossed content loses standing for the old scope. That standing does not cross the uncertified horizon. The interior cannot create replacement standing without certificate or new governance. But non-standing organization may remain. Such organization is not standing. It is residual coherence: the persistence of formerly governed role patterns after the standing relation that licensed them has been broken.

Residual coherence can be robust near the horizon and weaker at greater governance-depth. It can differ by role. Coordinates, foliations, and labels are weak because they were skin. Tensor-like load is stronger because it carried physical or structural content before the break. Source and metric roles may persist longer as residual load, but not as fully standing-bearing roles. Eventually the role-support for metric comparison, matter-energy, temporal order, and causality can become so thin that the ordinary language of inside, deeper, before, after, cause, field, and energy becomes a residual convenience rather than a stable descriptor.

## No local-dynamical discontinuity claim

The strong interpretation does not imply a local dynamical discontinuity at the horizon. A standing break is a governance-level transition, not an assertion that the Einstein projection fails at the crossing event. The local metric representative may remain regular at the boundary, and residual metric and matter coherence may be high immediately after crossing. The claim is not that the crossed structure instantly becomes chaos, that local matter is destroyed, or that the local field equation ceases to be the licensed metric projection. The claim is that local metric description is not by itself a certificate of same-scope standing transport for crossed content.

## Why the classical picture overstates the interior

Classical black-hole geometry extends a metric description into an interior. That extension is a useful licensed projection in the appropriate domain. But if the standing break has removed the governance support required for the crossed class, then the metric extension cannot be promoted into proof that the interior remains an ordinary physical interior. The metric representative is downstream. The governance relation is upstream. The strong interpretation respects that order.

A formal metric interior can continue as a residual projection even where same-scope standing-bearing interiority has failed. This is the sense in which the interior is not erased but residualized. It remains describable to the degree that residual coherence supports description. The paper classifies that degree structurally through role-indexed residual coherence and governance-depth.

## Kernel commitments

The AASC kernel is written

$$K_D = (Adm_D, Stand_D, Ref_D, Irr_D).$$

Here  $Adm_D$  distinguishes licensed from unlicensed continuation in a fixed domain,  $Stand_D$  records reusable standing status,  $Ref_D$  fixes what is being continued or identified, and  $Irr_D$  prevents same-

act repair after standing failure. These are not new physical substances. They are the governance requirements for non-degenerate construction or physical description.

### **Kernel deletion and interior-status loss**

The standingless-interior question exists only if the whole kernel is in force. Denying a component does not restore an ordinary interior. It makes the interior-status question ill typed.

Without  $Adm_D$ , there is no difference between a licensed interior role, a hidden selector, a coordinate continuation, a high-curvature label, a local tensor notation, and a post-hoc repair. A local metric expression or stress-energy ledger can then be promoted into ordinary interiority without passing any admissibility check. In that setting, the paper could no longer separate local AASC-GR writability from standing-bearing interior role licensing.

Without  $Stand_D$ , there is no standing break and therefore no controlled distinction between ordinary interiority, residual coherence, bookkeeping, and nullity. Residual coherence would be misread either as partial standing or as ignorance about a hidden ordinary interior. Without  $Ref_D$ , the crossed content  $x$ , the old standing class, the alleged interior target, and the residual role ancestry are indeterminate. The claim that “the same object” entered  $E_{\text{int}}$  would have no fixed target.

Without  $Irr_D$ , residual coherence can be repaired into fake standing after loss. A later dictionary, remnant label, final-state condition, quantum-gravity branch, or coordinate continuation could be paired with the old target and treated as if standing had been preserved all along.

Thus kernel deletion is not an interior rescue. It removes the basis on which one can distinguish same-scope standing-bearing interiority from residual metric projection, residual metrological coherence, hidden governance import, repair, or new scope. The theorem uses the kernel in exactly this way:  $Adm_D$  licenses or rejects an interior role,  $Stand_D$  marks the standing break,  $Ref_D$  fixes the crossed target, and  $Irr_D$  blocks repair of residual organization into preservation.

### **Standing is bivalent**

Standing is bivalent relative to a fixed scope:

$$Stand_D(x) \in \{0, 1\}.$$

A structure either has standing or does not. This is the first discipline of the paper. No theorem below says that standing becomes weak, fuzzy, partial, or probabilistic. Noise, approximation, probability, entropy, and partial record quality are downstream notions. Standing is governance-level eligibility for identity-preserving use.

### **Standing is not created internally**

The kernel also blocks internal creation of replacement standing. If standing is broken in a fixed scope, later assignment of equivalent standing is not preservation unless the preserving structure was already licensed. If the assignment is introduced after the break, it is repair. If it chooses an

unanchored continuation, it is selector import. If it changes the domain, it is scope change. If it produces an exterior recoverability map, it is a certificate.

This matters for the interior. If crossed standing is broken at the uncertified horizon, the interior does not spontaneously become a source of old-scope standing. A new interior physics may be declared, but then the claim has moved to a new governance scope or certificate relation. Without that, the old standing is gone.

### Governance admission and initial licensing

Standing is not generated by an interior process. It is instantiated by governance admission: a candidate reference class becomes standing-bearing when it is admitted into a domain governed by admissibility, standing, reference, and irreversibility. This first licensing is boundary-level, not an operation performed inside a pre-existing same-scope domain.

In schematic form,

$$K_D = (Adm_D, Stand_D, Ref_D, Irr_D) \Rightarrow D \text{ is a standing-bearing admissibility domain.}$$

But the reverse operation is not available internally. From the fact that  $x$  is not standing-bearing inside an already fixed domain  $D$ , it does not follow that some same-scope operation  $f_D$  can make it standing-bearing:

$$x \notin Stand_D \not\Rightarrow \exists f_D(x) \in Stand_D.$$

Such an operation would be a declared governance act, a certificate, a scope change, or repair. It is not ordinary interior continuation.

### Initial licensing is not repeatable by downstream boundaries

The initial licensing of standing downstream of the AMetric boundary is not a license for every later boundary to generate standing. The AMetric boundary is prior to metricity, comparison, causal order, scale, and physical interior roles. A downstream boundary that already presupposes those roles cannot repeat the AMetric licensing act in the same scope.

**Proposition (Initial Licensing Is Not Horizon-Repeatable).** The initial licensing of standing downstream of the AMetric boundary cannot be repeated by a downstream metric boundary unless that boundary is explicitly retyped as a new admissibility-generating boundary. Retyping it in that way changes scope.

**Proof.** Initial licensing occurs with the establishment of the governed domain  $D$ . A black-hole horizon appears only after  $D$  already contains metric/interior roles sufficient to define horizon structure. To use the horizon as a licensing boundary would require a new domain  $D'$  or a certificate relating  $D'$  to  $D$ . Therefore the horizon is not a same-scope repetition of the AMetric licensing boundary. QED.

## Standing break

A standing break occurs when a reference class ceases to have a licensed, re-fixable continuation under the declared witness and redescription structure. In the black-hole case, a threatened reference class  $r \in R_{\text{ext}}$  undergoes standing break when it lacks an admissible future image in the exterior witness algebra  $A_{\text{wit}}$ , the continuation relation  $Cont$ , a leakage route, a standing-transport certificate, or a non-sink quotient.

The standing break is not a statement about annihilation. It is a statement about role-status. The crossed content may remain representable, but not as standing-bearing continuation of the same reference class in the old scope.

## Standing versus residual coherence

The paper's central distinction is:

$$Stand_D(x) \in \{0, 1\}, \quad \text{ResCoh}_D^r(x) \text{ is role-indexed and gradable.}$$

Residual coherence is not standing. It is not a third standing value. It is non-standing persistence of role organization. Confusing the two produces the two main errors this paper avoids. If residual coherence is mistaken for standing, the interior becomes unqualified ordinary physical structure. If absence of standing is mistaken for nullity, the interior becomes nothing. The strong AASC position is neither: standing is absent, but residual coherence may remain.

## The horizon-isolated carrier

The black-hole reconstruction framework decomposes the horizon carrier as

$$X_H = E_{\text{ext}} \cup E_{\partial H} \cup E_{\text{int}}.$$

Here  $E_{\text{ext}}$  is the exterior witness envelope,  $E_{\partial H}$  is the boundary or horizon-trace envelope, and  $E_{\text{int}}$  is the horizon-isolated carrier relative to the declared reconstruction problem.

## The carrier is not first a metric container

The carrier  $E_{\text{int}}$  is not first a coordinate region. It is the sector on which standing transport for the crossed class is not licensed in the old scope. A metric projection may represent it as inside the horizon, nearer the singularity, or deeper in the geometry. Those are residual projections. The structural fact is that the old standing-transport assignment is not defined there.

## Standing transport failure

Let

$$T_H : X_H \rightarrow R_{\text{ext}}$$



be the horizon standing-transport assignment. In the uncertified permanent-horizon case,

$$E_{\text{int}} \cap \text{dom}(T_H) = \emptyset$$

for the threatened crossed reference class. This undefinedness is not ignorance and not temporary lack of data. It is non-membership in the old admissible operator domain.

### Standingless carrier corollary

**Theorem (Standingless Carrier Corollary).** In the uncertified permanent-horizon case, if no leakage, standing-transport certificate, non-sink quotient, or declared new governance scope is supplied, then  $E_{\text{int}}$  is a standingless carrier for the crossed content in the same-scope AASC sense.

**Proof.** The crossed content lacks an admissible image under the old exterior witness algebra and continuation relation. The horizon does not transport standing into  $E_{\text{int}}$  without certificate. The kernel does not permit same-scope creation of replacement standing inside  $E_{\text{int}}$ . Nor can the horizon itself license new standing, because the no-horizon-as-AMetric-reset theorem classifies it as a derivative metric boundary rather than an AMetric reset. Therefore the crossed content has no standing-bearing continuation in  $E_{\text{int}}$  under the old scope. What remains, if anything remains representable, cannot be standing. It is residual organization. QED.

### Non-nullity and residual description

A standingless carrier is not a null carrier. It may retain non-null residual load. It may still be represented by fields, curvature, stress, source, or matter descriptions over some residual range. Those descriptions are not full same-scope standing-bearing roles unless governance is reintroduced. This is why residual coherence is needed.

### The horizon is not an AMetric reset

The strongest interpretation requires a boundary distinction that is easy to miss. Standing may be licensed immediately downstream of the AMetric boundary, but a black-hole horizon is not such a boundary. The AMetric boundary is prior to metricity, comparison, scale, causal order, and physical interior roles. A black-hole horizon is already a licensed metric-interior structure. Its own intelligibility depends on metric comparison, causal organization, exterior/interior role typing, and standing-governed reconstruction. It cannot generate the governance conditions it presupposes.

### Admissibility-generating and derivative metric boundaries

**Definition (Admissibility-generating boundary).** An admissibility-generating boundary is a pre-metric boundary condition at which a governed domain first becomes possible. It does not rely on an already licensed metric, causal, source, or witness structure. The AMetric boundary is of this type.

**Definition (Derivative metric boundary).** A derivative metric boundary is a boundary structure defined inside an already licensed metric/interior projection. Its intelligibility depends on prior

governance, comparison, and standing-bearing description. It can obstruct, filter, or redirect standing transport, but it cannot generate the governance conditions that make standing possible.

Thus

$$B_{\text{AMetric}} = \text{admissibility-generating boundary},$$

whereas

$$H_{\text{BH}} = \text{derivative metric boundary}.$$

| Feature                               | AMetric boundary                     | Black-hole horizon                                  |
|---------------------------------------|--------------------------------------|-----------------------------------------------------|
| Dependency level                      | upstream of metricity                | downstream of metricity                             |
| Role                                  | licenses the first admissible regime | obstructs, filters, or redirects standing transport |
| Requires metric projection            | no                                   | yes                                                 |
| Requires standing-governed comparison | no; it is prior to such comparison   | yes                                                 |
| Can license standing                  | yes, as governance admission         | not in the same scope                               |
| If treated as licensing boundary      | ordinary AASC origin                 | new scope, certificate, or hidden import            |

### Horizon identity depends on prior metric governance

**Lemma (Horizon Identity Depends on Prior Metric Governance).** A black-hole horizon, considered as the same-scope horizon of a licensed black-hole metric-interior projection, is not identified prior to governance. Its role requires at least a metric or metric-projection structure, causal or exterior/interior separation, comparison sufficient to distinguish inaccessible or one-way regions, witness or reconstruction typing for the exterior/interior relation, and standing-governed continuation sufficient to classify the boundary as a horizon rather than arbitrary skin. Schematically,

$$H_{\text{BH}} \Rightarrow \text{Metric}_D \wedge \text{Causal}_D \wedge \text{Compare}_D \wedge \text{WitnessTyping}_D.$$

Since metricity, causal typing, comparison, and witness typing are downstream of governance in AASC,  $H_{\text{BH}}$  cannot be the same-scope origin of the governance conditions that make it identifiable.

**Proof.** A horizon is not a bare mark. In the classical/event-horizon role it is fixed by metric and causal structure; in trapping or apparent-horizon roles it is fixed by metric, foliation, expansion, or related geometric data; in reconstruction roles it is fixed by exterior/interior witness typing and continuation burdens. Each route already presupposes licensed comparison and role-governed description. A boundary whose identity depends on those structures cannot, in the same scope, generate those structures as if it were an AMetric licensing boundary. Retyping such a boundary as

fundamental is possible only by declaring a new theorem object, new governance scope, or certificate. QED.

### No Horizon-as-AMetric-Reset Theorem

**Theorem (No Horizon-as-AMetric-Reset, dependency form).** Let  $H_{\text{BH}}$  be a black-hole horizon considered as the same-scope horizon within a licensed metric-interior projection  $D$ . Since  $H_{\text{BH}}$  depends on metric comparison, causal typing, and exterior/interior reconstruction roles, it cannot function as the same-scope AMetric licensing boundary for those roles. If horizon crossing is claimed to license new interior standing, that claim is exactly one of: declared new governance scope, certificate reconnecting to the old standing class, richer-scope physical theory, hidden governance import, post-hoc repair, or bookkeeping. It is not same-scope standing creation by the horizon itself.

**Proof.** The AMetric boundary is prior to metricity, scale, causal order, and comparison. It can license the first admissibility-bearing regime because it does not presuppose the metric/interior structure it licenses. A black-hole horizon has the opposite dependency order. By the horizon-identity dependency lemma, the same-scope horizon is identifiable only through metric comparison, causal typing, exterior/interior role typing, and a declared witness or reconstruction scope. These are downstream of governance. A structure that depends on standing-governed comparison cannot same-scope license the governance that makes that comparison meaningful. Retyping the horizon as an admissibility-generating boundary is not same-scope horizon crossing; it introduces a new governance boundary  $D'$ , a certificate into  $D$ , a richer physical theory, hidden governance import, repair, or bookkeeping. Therefore a black-hole horizon, as the same-scope horizon of the existing metric-interior projection, is not an AMetric reset. QED.

The theorem permits richer theories; it only blocks same-scope horizon-as-origin. The following table records the classification.

| Proposal                                    | AASC classification                        | Effect on the no-reset theorem                  |
|---------------------------------------------|--------------------------------------------|-------------------------------------------------|
| Classical or event horizon                  | derivative metric boundary                 | cannot license standing in the same scope       |
| Holographic screen                          | possible certificate or richer governance  | not same-scope reset unless declared and mapped |
| Island or quantum extremal surface boundary | reconstruction certificate or richer scope | must be declared as a new reconstruction object |
| Quantum-gravity transition surface          | new governance scope $D'$                  | scope change, not ordinary horizon crossing     |
| Interior fundamental boundary               | new admissibility-generating boundary      | new theorem object, not old-scope horizon       |
| Undeclared horizon-origin                   | hidden governance import                   | inadmissible in the old scope                   |
| Later recovery of standing                  | repair unless certified                    | not same-scope preservation                     |

### Post-metric governance deficiency

The residual-coherence interior is not a return to the AMetric boundary. It is post-metric, not pre-metric. Metric roles were once licensed in the exterior and in the classical representation, but

after the standing break they have residual metric-projection status unless re-governed. The horizon is therefore not a new origin of standing. It is a downstream obstruction internal to a licensed regime.

This distinction controls the whole manuscript. Residual coherence becomes the default status only after three routes have been excluded: standing does not cross; standing is not created internally; and the horizon is not an AMetric reset. If a new AMetric-like boundary is posited inside the black-hole regime, then the theorem object changes from  $D$  to  $D'$ . That may be a legitimate richer theory, but it is not same-scope horizon crossing.

## GR in AASC: local metric projection versus standing transport

This paper presupposes the AASC Einstein projection. It does not deny the local validity of the licensed metric projection, and it does not claim a local dynamical discontinuity at horizon crossing. Its target is different: whether the same-scope metric projection, at an uncertified permanent-horizon endpoint, supplies standing transport, standing recovery, or AMetric re-licensing for crossed content. The answer is negative unless certificate, leakage, non-sink quotient, declared high-curvature or effective load, or richer governance is supplied.

The AASC Einstein projection has the schematic form

$$\Pi_g(A_G) = (M, g_{\mu\nu}), \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

In that projection, the metric equation is the unique minimal first-response same-scope metric form of the AASC gravity role. It remains a licensed local metric projection. But a licensed metric projection is not the same object as a standing-transport certificate. A metric continuation can be locally valid while a crossed reference class lacks standing-preserving continuation in the old scope.

## Same-scope AASC-GR horizon

**Definition (Same-scope AASC-GR horizon).** In this paper, a black-hole horizon means a horizon role inside an already licensed same-scope AASC metric projection: an event horizon, trapped-region boundary, or exterior reconstruction boundary as treated within that projection. It does not include a separately declared holographic screen, quantum extremal surface, island boundary, quantum-gravity transition surface, or fundamental admissibility-generating boundary unless those are introduced as certificate or new governance structures.

| Horizon-like structure       | AASC status in this paper                                                              |
|------------------------------|----------------------------------------------------------------------------------------|
| Classical event horizon      | derivative metric boundary; no same-scope AMetric reset                                |
| Apparent or trapping horizon | projection-internal horizon role; scope details must be declared                       |
| Isolated horizon             | boundary regime inside metric projection; certificate needed for standing preservation |

| Horizon-like structure                      | AASC status in this paper                                         |
|---------------------------------------------|-------------------------------------------------------------------|
| Holographic screen                          | possible certificate or richer governance if declared             |
| Quantum extremal surface or island boundary | reconstruction certificate or richer scope, not same-scope reset  |
| Quantum-gravity transition surface          | new governance scope $D'$                                         |
| Fundamental interior boundary               | admissibility-generating only if explicitly declared as new scope |
| Undeclared horizon-as-origin                | hidden governance import or repair                                |

### AASC-GR compatibility theorem

**Theorem (AASC-GR Compatibility of the Standingless-Interior Result).** Let  $\Pi_g(A_G) = (M, g_{\mu\nu})$  be the faithful AASC Einstein projection satisfying

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

The standingless-interior theorem does not deny the local validity of this metric projection. It states only that, at an uncertified permanent-horizon endpoint, the metric projection cannot by itself supply standing transport, standing recovery, or AMetric re-licensing for crossed content. Any such preservation requires certificate, leakage, non-sink quotient, declared high-curvature or effective load, or richer governance.

**Proof.** The AASC Einstein projection licenses local metric dynamics as a faithful projection of the AASC gravity role. That projection is downstream of standing, reference, quotient identity, and continuation. A downstream projection cannot retro-authorize the standing of the regime that licensed it. A black-hole horizon is a structure inside that projection. Therefore the horizon cannot use the metric projection to create new same-scope standing. If standing preservation is claimed at the endpoint, the claim must supply certificate, leakage, non-sink quotient, declared high-curvature or effective load, or richer governance. Hence the standingless-interior theorem is compatible with AASC-GR. QED.

**Corollary (Local Metric Validity Is Not Standing Transport).** The local validity of the AASC Einstein projection near or at a horizon does not imply that crossed content remains standing-bearing for the exterior same-scope reconstruction class:

$$AASC-GR_D(H) \not\Rightarrow StandTransport_D(x, H).$$

The Einstein projection can license local curvature/source response while still failing to supply the certificate required to preserve the standing of crossed content.

## Metric continuation and standing continuation

**Definition (Metric continuation).** Metric continuation is continuation of the metric projection according to its licensed local curvature/source equations.

**Definition (Standing continuation).** Standing continuation is continuation of a standing-bearing reference class under admissible reconstruction, witness, certificate, or quotient-preserving transport.

The two are not equivalent:

$$MetricContinuation_D(x) \not\equiv StandingContinuation_D(x).$$

This distinction is decisive for the interior theorem. A classical extension or local metric evolution may remain formally meaningful while standing transport for crossed content is absent.

## No metric-projection self-certification

**Theorem (No Metric-Projection Self-Certification).** A faithful metric projection can express standing-bearing dynamics only after the physical interior is licensed. It cannot certify the preservation, recovery, or re-licensing of standing for content whose standing is threatened at a horizon endpoint unless a certificate, leakage channel, non-sink quotient, declared high-curvature or effective load, or richer governance is supplied.

**Proof.** Metric projection is downstream of AASC standing and role licensing. A downstream projection cannot retro-authorize the standing of its own licensing regime. Horizon crossing threatens standing transport for crossed content. If the metric projection itself is used as the certificate, that is metric backflow. Metric backflow is excluded by the AASC projection order. Therefore independent endpoint structure is required. QED.

## No same-scope interior governance

The strongest version of the paper requires a direct answer to the obvious objection: why can the interior not simply have its own standing? It can, but only if a governance structure is declared. A declared interior governance is new scope unless it is already a certificate that connects to the old standing class. The uncertified same-scope case excludes both.

## Interior governance structure

A same-scope interior governance structure for crossed content would be a tuple

$$G_{\text{int}}^D = (Adm_{\text{int}}, Stand_{\text{int}}, Ref_{\text{int}}, Irr_{\text{int}}, W_{\text{int}}, Cont_{\text{int}})$$

that preserves the crossed content as the same standing-bearing target, remains in the old domain  $D$ , supplies reference fixation inside  $E_{\text{int}}$ , does not change scope, and does not function as a hidden certificate or post-hoc repair. If any of these conditions fail, the structure is not same-scope interior governance in the old AASC sense.

A declared new domain  $D'$  may support an interior governance

$$G_{\text{int}}^{D'} = (\text{Adm}'_{\text{int}}, \text{Stand}'_{\text{int}}, \text{Ref}'_{\text{int}}, \text{Irr}'_{\text{int}}, W'_{\text{int}}, \text{Cont}'_{\text{int}}),$$

but that is new governance. If it maps back to the old crossed reference class, it is a certificate. If it does not, it may be an interior theory but does not preserve old standing.

### No Same-Scope Interior Governance Theorem

**Theorem (No Same-Scope Interior Governance).** Let  $D$  be the original admissibility-governed scope, let  $E_{\text{int}}$  be the horizon-isolated carrier reached after an uncertified permanent-horizon transition, let  $H_{\text{BH}}$  be the derivative metric horizon boundary, and let  $x$  be crossed content whose pre-horizon standing has been broken. If no certificate, leakage route, non-sink quotient, or declared new interior governance scope is supplied, then  $E_{\text{int}}$  cannot instantiate a same-scope standing-bearing physical interior for  $x$ .

Equivalently,

$$\text{Stand}_D(x) = 0 \wedge \neg \text{Cert}_H(x) \wedge \neg \text{NewGov}(D') \Rightarrow \neg \text{RPI}_D(E_{\text{int}}, x).$$

**Proof.** A realized physical interior in the same scope requires the kernel roles for the content under consideration: admissibility, standing, reference, and irreversibility. By hypothesis,  $x$  has lost standing in  $D$ . Standing is not transported through the uncertified horizon. Standing is not created internally in  $D$ . Nor can  $H_{\text{BH}}$  itself license standing, because it is a derivative metric boundary and not an AMetric reset. Therefore a claim that  $E_{\text{int}}$  supplies standing for  $x$  must introduce some additional structure. If that structure maps  $x$  back to the old standing class, it is a certificate. If it supports a new domain, it is new governance scope. If it is undeclared, it is hidden governance import. If it is added after loss as preservation, it is repair. All of these are excluded by hypothesis. Thus  $E_{\text{int}}$  is not a same-scope standing-bearing physical interior for  $x$ . QED.

### Strong standingless interior thesis

The theorem yields the strong interpretation:

In the uncertified permanent-horizon case, absent certificate or declared new governance, the horizon-isolated carrier is not a same-scope standing-bearing physical interior. It is a residual-coherence carrier.

This is stronger than saying merely that the exterior cannot reconstruct the interior. It says that the ordinary physical interior role architecture is not inherited by crossed content after standing loss. The interior may be re-governed by a larger theory, but without such re-governance it is not unqualified ordinary physical interiority.

### Re-governance classification

Any claim that  $E_{\text{int}}$  has standing after the break falls into one of the following:

| Claim                                   | Classification                                         |
|-----------------------------------------|--------------------------------------------------------|
| Old standing is preserved inside        | Certificate required; otherwise false same-scope claim |
| New standing exists inside              | Declared new governance scope                          |
| New standing maps to old class          | Certificate or re-coupling map                         |
| New standing does not map to old class  | Disconnected interior theory                           |
| Standing is silently assumed            | Hidden governance import                               |
| Standing is assigned after loss         | Repair                                                 |
| A label is added without witness change | Bookkeeping                                            |

No seventh route is available inside the fixed same-scope problem.

### Why the strong theorem is still conditional

The No Same-Scope Interior Governance Theorem is deliberately strong, but it is not unconditional ontology. It has three gates. First, it assumes the uncertified permanent-horizon case. If a leakage route, radiation witness, standing-transport certificate, or non-sink quotient is supplied, the theorem’s antecedent fails. Second, it assumes the old same-scope governance domain. If a new domain is declared, the question becomes whether that new domain is admissible and whether it maps to the old crossed class. Third, it assumes the RPI dependency order: metricity, matter, energy, time, and causality are not primitive carriers of their own admissibility.

These gates matter because the paper is not trying to defeat every possible quantum-gravity account of the interior. It is trying to prevent an unlicensed inference: because the metric extension can be written, ordinary physical interiority continues. That inference reverses the dependency order. A metric continuation is a representation inside a licensed regime; it is not itself the governance structure that licenses matter, reference, standing, and continuation.

The strong theorem therefore has the following form:

$$\begin{aligned} &\text{uncertified horizon break} + \text{no AMetric reset} + \text{no re-governance} \\ &+ \text{RPI dependency} \Rightarrow \text{residual-coherence carrier.} \end{aligned}$$

It is a theorem about what cannot be silently assumed after standing loss. It is not a denial that a declared stronger theory may change the result. A declared stronger theory is exactly what the theorem demands.

### The failure of ordinary interior inheritance

The ordinary inheritance picture says that the crossed system remains the same physical system inside the horizon, with the same ordinary matter, energy, metric, time, and causal roles, merely inaccessible to exterior observers. That picture is not kernel-faithful in the uncertified case. If standing does not cross and is not recreated, there is no old-scope standing-bearing target to which the ordinary interior roles can attach. The old physical identity has no standing-bearing continuation in the carrier. Therefore ordinary interior inheritance is blocked.



The block is not epistemic. It does not say that exterior observers lack knowledge while a complete interior fact remains. It says that the old governance relation does not extend. A structure that cannot inherit standing may still retain residual coherence, but it cannot be treated as the same standing-bearing physical structure without additional governance.

### The exact location of the substantive step

The substantive step is not the claim that black-hole interiors are strange. The substantive step is the claim that ordinary physical role-status is not primitive. If metricity, matter, energy, time, and causality were primitive, then the interior could simply keep them. Under AASC they are not primitive. They are roles in a standing-bearing interior architecture. Once standing is absent and no new governance is declared, those roles cannot remain fully licensed.

This is why the paper's conclusion is stronger than exterior inaccessibility but weaker than annihilation. The interior is not unqualified ordinary physics. It is not nullity. It is post-standing residual organization.

## Part II - Residual Coherence

### Residual coherence as theorem object

The No Same-Scope Interior Governance Theorem forces a third category between ordinary standing-bearing interior and nullity. If crossed content is not standing-bearing in the old scope but remains non-nullly representable, the remaining structure must be classified. This paper calls that structure residual coherence.

#### Definition

Let  $\mathcal{R}_D$  be the role family in the relevant physical scope:

$$\mathcal{R}_D = \{skin, record, gauge, tensor, metric, measurement, field, particle, source, time, causal\}.$$

For  $r \in \mathcal{R}_D$ , the expression

$$\text{ResCoh}_D^r(x)$$

means role-indexed residual coherence: the non-standing persistence of role organization of type  $r$  after standing has been broken for  $x$ . It is not whatever remains after standing loss. It is licensed only when six conditions obtain:

1. **Pre-break role ancestry:**  $x$  participated in role family  $r$  before the standing break.
2. **Standing loss:** after the uncertified transition,  $\text{Stand}_D(x) = 0$ .
3. **No re-governance:** no certificate, leakage route, non-sink quotient, or new governance scope is supplied.
4. **Post-break non-null representation:** some post-break representation remains describable.

5. **Retained organization:** the representation retains structure traceable to pre-break role  $r$ .
6. **Failure of full licensing:** the retained organization cannot satisfy the full standing-bearing role conditions of  $r$ .

The superscript is essential. There is no single undifferentiated coherence. Skin coherence, tensor-load coherence, metric-comparison coherence, particle-identity coherence, source-ledger coherence, and causal-dependence coherence fail differently because they depended on different role supports before the break.

**Lemma (Residual Coherence Requires Role Ancestry).** If  $x$  lacks pre-break role ancestry of type  $r$ , then residual coherence of type  $r$  cannot be asserted merely because some post-break representation remains:

$$\neg \text{Ancestry}_D^r(x) \Rightarrow \neg \text{ResCoh}_D^r(x).$$

**Proof.** Residual coherence is inherited organization. Without pre-break role ancestry there is no role organization of type  $r$  to inherit. A remaining representation may be another residual role, a new governance claim, hidden import, or exhaustion, but it is not  $\text{ResCoh}_D^r(x)$ . QED.

### Residual coherence is not standing

The definition includes the exclusion:

$$\text{ResCoh}_D^r(x) \not\Rightarrow \text{Stand}_D(x) = 1.$$

Residual coherence cannot license continuation, re-fix reference, repair standing, or supply a certificate. It can explain why a formerly standing structure remains approximately describable after the break. It cannot make that structure standing-bearing in the old scope.

### Residual persistence preorder

Define a structural comparison:

$$\text{ResCoh}_D^r(x) \preceq \text{ResCoh}_D^s(y)$$

when the residual role organization of type  $r$  has no more persistence under governance-depth than the residual role organization of type  $s$ . This is not a physical time rate and not a numerical measure. It is a fixed-domain comparison of role-support persistence.

A role-load relation can be written schematically as

$$\text{Load}(r) < \text{Load}(s) \Rightarrow \text{ResCoh}_D^r \prec \text{ResCoh}_D^s$$

under comparable governance-deficiency conditions. The relation is structural. It says that roles that carried less load before the standing break have weaker residual persistence afterward.

## Residual-Coherence Classification Theorem

**Theorem (Residual-Coherence Classification).** If  $x$  is post-break non-nullly represented, has pre-break role ancestry in some  $r \in \mathcal{R}_D$ , lacks standing, and no certificate or new governance is supplied, then any admissible description of  $x$ 's retained role organization is residual coherence rather than standing-bearing role continuation.

Formally,

$$NonNullRep(x) \wedge Ancestry_D^r(x) \wedge Stand_D(x) = 0 \wedge \neg NewGov \wedge \neg Cert \Rightarrow ResCoh_D^r(x).$$

**Proof.** Non-null representation blocks the classification as immediate nullity. Pre-break ancestry supplies role organization capable of leaving a trace. Standing loss blocks classification as standing-bearing continuation. The absence of new governance or certificate blocks re-licensing. If retained organization remains traceable to role  $r$  while full role licensing fails, the only same-scope classification is residual coherence of type  $r$ . QED.

Residual coherence is therefore not the final substrate of the interior. When role-indexed retained organization is absent, residual coherence no longer applies. The appropriate classification is exhaustion or nullity relative to the role vocabulary, not a minimal surviving stuff:

$$\forall r \in \mathcal{R}_D \neg ResCoh_D^r(x) \Rightarrow \text{no role-indexed residual description is licensed.}$$

## Robustness of the pre-break structure

Residual coherence depends on the robustness of the original role description. A structure whose pre-break standing was strongly anchored by boundary records, tensor load, gauge-invariant content, source/deformation response, or redundant witness relations has more residual organization to lose. A weak label, chart choice, or representative convention has little residual organization. This is why the coherence of a strongly organized crossed system may remain viable at greater governance-depth in the carrier than the coherence of a weakly anchored description.

## Formal properties of residual coherence

The role-indexed notation is intentionally minimal. Residual coherence is not introduced as a new physical field, scalar, entropy, or probability measure. It is a structural status assigned when three conditions hold:

1. the target no longer has standing in the original scope;
2. the target remains non-nullly representable in some role-language;
3. the representation inherits organization from a formerly governed role.

Thus  $ResCoh_D^r(x)$  is a predicate family with an optional comparison relation. It is not a quantity that must be conserved. It is not an energy. It is not a hidden information reservoir. It is a way of saying that a non-standing structure still retains some role-pattern inherited from a standing-bearing regime.

### Residual coherence and successor support

Residual coherence can be linked to successor support. Let  $Succ_D^r(x)$  be the residual successor neighborhood for role family  $r$ , meaning the class of role-continuations that preserve enough residual organization for that role to remain describable. Then one may read

$$\text{ResCoh}_D^r(x) \preceq \text{ResCoh}_D^s(y)$$

as a statement about relative successor support: the residual successors of  $r$  are no richer than those of  $s$ , up to residual-equivalent redescription. This is not a claim about physical time evolution. It is a comparison of how much structural support remains for a role to be continued as a residual description.

### Residual coherence and non-nullity

Residual coherence presupposes non-null representability. If no structure remains, there is no residual coherence to discuss. But non-nullity alone does not provide residual coherence. A random label or invented interior coordinate does not count. The residual structure must inherit role organization from the pre-break regime. This avoids turning residual coherence into arbitrary projection.

### Residual coherence and approximation

Residual coherence can explain why approximate classical descriptions may remain useful. If residual metric coherence is high, a metric continuation can remain an excellent approximation. If residual source coherence is high, matter-energy descriptions can remain serviceable. The approximation is not evidence that standing persisted. It is evidence that residual coherence remains high in the relevant role family.

### Residual coherence and certificates

A certificate changes the status of the situation. A certified boundary reconstruction, holographic map, island reconstruction, remnant map, or leakage channel may re-establish standing relations. Once it does, the relevant content is no longer merely residual. It has been re-governed. The residual-coherence theorem is a theorem about the uncertified case.

### Role ancestry and residual persistence

Role ancestry is the pre-break structural load carried by a role. It is not a psychological or epistemic ranking. It asks how much standing-bearing work the role performed before standing was broken.

### Role-support dependency order

The order displayed below is heuristic and dependency-relative. The formal claim is partial-order support dependence, not a universal scalar ranking of roles. Gauge presentation, record structure, source response, and metric comparison may occupy different support positions in different declared scopes.

A schematic role-support dependency order is

$$\begin{aligned} &Skin < Record/Bookkeeping < GaugePresentation \\ &< TensorLoad < Source/MetricLoad. \end{aligned}$$

The displayed order is heuristic and dependency-relative, not a universal scalar ranking of roles. It indicates the typical order of residual support after governance loss. Skin is weakest because it is presentation over role content. Tensor load is stronger because it carried invariant or compatibility content. Source and metric load are strong because they organize response, comparison, and continuation.

No role-support order contains a final primitive remainder. The exhaustion boundary is not a role-load family. It is the point at which no role-indexed residual coherence remains.

A role family  $r$  is said to have stronger residual persistence than  $s$  only when four basis conditions hold:  $r$  carried more pre-break structural load,  $r$  is less dependent on representative skin,  $r$  has more inter-role support in the realized physical interior chain, and  $r$  can retain organization under thinner residual successor support. These conditions justify the dependency order; they are not a numerical measure.

### Role-Ancestry Persistence Theorem

**Theorem (Role-Ancestry Persistence).** In a standingless carrier, residual coherence persists according to pre-break role load. If  $Load(r) < Load(s)$ , then under comparable governance-depth  $ResCoh_D^r$  has weaker residual persistence than  $ResCoh_D^s$ .

**Proof.** Residual coherence is inherited from pre-break role organization. A role with less pre-break load contributed less independent structure to the governed domain. Once the governance relation is broken, such a role has less non-standing organization to persist. A role with greater load carried more invariant, tensor-like, source-like, or metric-comparison organization. That organization can remain as residual coherence even when standing is absent. Therefore residual persistence follows role ancestry. QED.

### Coherence persistence is not standing persistence

Robustness of a role before the break does not preserve standing through the break. The robust structure has  $Stand_D = 0$  in the old scope if the crossing is uncertified. What robustness affects is the degree of residual coherence after standing loss. This is the formal home of the intuition that a strongly organized object may remain approximately coherent at greater governance-depth than a weak representative label.

## Part III - Skin, Tensor, and Role-Support Order

### Skin-level support loss

Skin is presentation without independent standing-bearing load. Coordinates, foliations, labels, chart choices, representative radial depth, local names, and verbal descriptions are skin when their

variation changes no tensor or anchor content in the governed domain. Skin is not false; it is often indispensable for calculation. But it does not carry independent role support.

### Skin after standing break

After a standing break, skin has little residual support because it depended on role structures that no longer have standing. A coordinate label may still be written. A foliation may still be formally selected. A chart-defined radial parameter may still appear in an equation. But the support that made those descriptions stable now has residual status. Skin loses support at shallow governance-depth.

### Skin-First Residual Support Failure Theorem

**Theorem (Skin-First Residual Support Failure).** Let  $r_{skin}$  be a skin role and  $r_{ten}$  a tensor-like role in the pre-break domain. Then

$$Load(r_{skin}) < Load(r_{ten}) \Rightarrow \text{ResCoh}_D^{r_{skin}} \prec \text{ResCoh}_D^{r_{ten}}$$

under comparable governance-depth.

**Proof.** Skin carried no independent tensor load before the break. Its stability depended on the standing of the roles it represented. Tensor-like roles carried load-bearing content before the break. Once standing is broken, skin has only derivative support, while tensor-like roles have residual organization inherited from their previous load. Therefore skin has weaker residual persistence. QED.

### Consequences for interior description

The first structures to become unreliable in a standingless interior are skin-like structures: coordinate depth, foliation-dependent simultaneity, chart labels, representative singularity loci, and unanchored gauge choices. This does not mean tensor-like structure has full standing. It means skin loses residual support before tensor-like load does.

### Near-horizon local metric regularity and skin-level support loss

Near the horizon, residual tensor and metric coherence may be high enough that ordinary local descriptions remain serviceable. The standing break does not require a local dynamical discontinuity. But skin-level descriptions are already vulnerable because the governance support that stabilized their role has been removed. The fact that local fields or stress may remain coherent does not imply that all representative descriptions retain equal status.

### Tensor-residual persistence

Tensor-like load includes curvature response, field stress, electromagnetic response, angular-momentum response, deformation/source response, bounded transport pattern, and metric-facing source ledger. Such structures carried load-bearing role content before the standing break. Their alteration would have changed the governed structure, not merely its presentation.

## Tensor-like persistence without tensor standing

After the standing break, tensor-like load may remain organized. It may continue to support approximate curvature response, field stress, source deformation, or angular momentum response in the residual carrier. But it does not remain standing-bearing tensor content unless a certificate or new governance structure licenses it.

The correct implication is:

$$\text{ResCoh}_D^{\text{tensor}}(x) > 0 \not\Rightarrow \text{Stand}_D(x) = 1.$$

## Tensor-Residual Persistence Theorem

**Theorem (Tensor-Residual Persistence).** Tensor-like load has greater residual persistence than skin because it carried pre-break load-bearing structure. However, residual tensor persistence is not tensor standing.

**Proof.** Tensor-like roles carried invariant or compatibility content in the governed domain. When standing is broken, this content does not thereby become standing-bearing in the interior. But the organization it had before the break can persist as residual structure. Skin lacks such load and therefore has weaker residual persistence. Since the kernel makes standing bivalent and no certificate is supplied, residual tensor organization cannot count as standing. QED.

## Examples of tensor residuals

Curvature response may remain formally describable after crossing, but its status is residual metric load. Field stress may persist as a pattern, but no longer with full field-carrier standing. Electromagnetic response may retain residual organization, but not boundary-fixed charge standing unless re-governed. Angular momentum response may persist as residual rotational load, but not as a fully licensed asymptotic or horizon-thermodynamic role inside the old scope.

## Role-support matrix

The following matrix is a role-support dependency taxonomy, not a flat ontology and not a strict total hierarchy. The role families are grouped by how their support depends on governed roles. Gauge presentation, for example, may be skin in one context and tied to load-bearing structure in another. Residual coherence is therefore assessed role-by-role, not by assigning each item to a universal ladder.

The following matrix is central to the paper. It does not describe dynamical decay in proper time. It describes role-support persistence under governance-depth.

| Role family | Pre-break status            | Post-break residual      |                                     |
|-------------|-----------------------------|--------------------------|-------------------------------------|
|             |                             | status                   | Principal failure mode              |
| Skin        | representative presentation | weakest residual support | chart/label/foliation unreliability |

| Role family        | Pre-break status                        | Post-break residual status  | Principal failure mode          |
|--------------------|-----------------------------------------|-----------------------------|---------------------------------|
| Gauge presentation | boundary-fixed presentation equivalence | ambiguous unless re-fixed   | gauge artifact                  |
| Record/information | witness-dependent standing              | residual bookkeeping only   | encoding/re-fixability loss     |
| Tensor-like load   | load-bearing role                       | stronger residual coherence | eventual support exhaustion     |
| Metric comparison  | licensed comparison role                | residual metricity          | depth/curvature instability     |
| Field carrier      | persistence mediation                   | residual carrier pattern    | carrier identity diffusion      |
| Particle class     | localized persistence class             | residual localization       | class identity loss             |
| Mass/inertia       | deformation-response role               | residual response           | source ambiguity                |
| Energy/source      | transport/source ledger                 | residual load               | conservation-ledger instability |
| Time               | retained consequence ordering           | residual order              | before/after instability        |
| Causality          | witnessed dependence                    | residual dependency         | causal ambiguity                |

This matrix should not be read as a temporal sequence. It is an order of role-support vulnerability. Some physical configurations may retain a high degree of residual coherence in several role families at once; others may lose coherence quickly. The structural rule is that lower-load roles cannot outlast the support of the higher-load roles they merely presented.

### Worked example: a structured probe after standing break

Consider a structured probe prepared outside the horizon. Before crossing, it has standing in the exterior scope because its identity is anchored by preparation records, detector correlations, and a declared redescription class. It may also have robust internal organization: bound constituents, field configuration, charge response, mass/inertia profile, and possibly an entangled reference relation. In the exterior domain these features are not merely skin; they contribute to the standing-bearing identity of the probe.

In an uncertified permanent-horizon transition, the probe's exterior standing is broken. The reference class is not transported into  $E_{\text{int}}$  as standing-bearing continuation. The interior does not create replacement standing. The probe therefore does not enter as an ordinary same-scope physical object in the strong AASC sense.

Nevertheless, the probe need not become null. Its bound structure may remain residually coherent. Its field configuration may remain approximately organized. Its source load may continue to support



residual deformation response. Its local metric description may still be usable near the horizon. The ordered status is:

1. standing of the crossed class is absent;
2. skin descriptions of the probe are weakest;
3. tensor-like field/source load remains stronger;
4. particle identity persists only residually;
5. mass/inertia persists as residual deformation response;
6. time and causality retain descriptive force only while residual order and dependency remain coherent.

This example captures the intuitive local-regularity point without abandoning the strong theorem. The probe may behave nearly normally near the horizon because residual coherence is high. But that ordinary behavior is no longer fully standing-bearing same-scope interiority. It is high residual coherence after standing break.

**Constrained handoff map for the probe.** The example has the following dependency shape. The arrows are status handoffs, not dynamical stages in proper time.

| Arc stage                                  | Status of the structured probe                                                                                                 |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Exterior standing-bearing probe            | Preparation records, witness relations, and declared re-description rules fix the probe as a standing-bearing exterior class.  |
| Horizon standing break                     | In the uncertified permanent-horizon case the old standing-transport map is undefined on the horizon-isolated carrier.         |
| No transport and no reset                  | The crossed class is neither transported as old standing nor relicensed by the horizon as an AMetric boundary.                 |
| Residual skin                              | Names, coordinates, and foliation-dependent descriptions have weakest residual support.                                        |
| Residual tensor/source/metric organization | Bound structure, field/source load, and local metric organization may remain strongly representable without becoming standing. |
| Residual metric-support exhaustion         | If residual metric-comparison successor support becomes empty, the endpoint antecedent exported to the homology paper is met.  |

### Worked example: representative depth versus governance-depth

In a classical chart one may label an interior curve by decreasing radial coordinate. The label may say that the system is deeper inside the black hole. Early after crossing, this description may track residual metric comparison well. The chart has residual coherence because the metric representation remains coherent enough.

As governance-depth increases, the residual successor support for metric comparison thins. The radial coordinate may remain formally present, but the role it played as stable comparison weakens. Eventually the coordinate may continue as skin while no longer tracking stable metric comparison.

At that point the phrase “deeper” is not false in the ordinary coordinate sense; it is unsupported as a standing-compatible or residual-coherent physical descriptor.

This example clarifies why the paper says depth loses meaning. It does not claim that a coordinate cannot be written. It claims that the coordinate’s role-support as metric comparison fails.

## Part IV - Metricity, Depth, and the Failure of Interior Geometry

### Metricity depends on governance

Metricity is not primitive in AASC. Metric description is licensed comparison, not a pre-governance container. A metric expression compares, orders, or relates. Such comparison requires reference stability, admissibility, standing, and redescription discipline. A curvature tensor, causal cone, proper distance, or radial coordinate is meaningful only after the metric role has been licensed.

### Metric Governance-Dependence Theorem

**Theorem (Metric Governance-Dependence).** In a standingless same-scope carrier without new governance, metric structures, where retained for crossed content, have residual projection status rather than fully licensed same-scope metric role status.

Formally,

$$Stand_D = 0 \wedge \neg NewGov(D') \Rightarrow Metric_D(E_{int}) \text{ is residual, not fully licensed.}$$

**Proof.** Metricity is quotient-stable comparison. Quotient-stable comparison requires reference and standing under permitted redescription. In the standingless carrier, standing is absent for the crossed content in the old scope. No new governance is declared. Therefore the metric role cannot remain fully licensed for that content. Any metric description that remains is residual projection. Because measurement is a governed use of metric comparison, loss of standing for metric comparison propagates to measurement-tied roles unless a separate metrological governance bridge is declared. QED.

### Consequences for curvature

Curvature scalars, curvature tensors, trapped-surface descriptions, and singularity loci may remain formally writable. But their role-status changes. They no longer automatically have the standing of fully licensed metric tensor content. They are residual metric descriptions whose stability depends on residual coherence. As governance-depth increases, those descriptions become less stable as invariant role carriers.

### Consequences for local continuation

Geodesic continuation, local evolution, and causal propagation depend on metric comparison and local domains. If metricity becomes residual, local continuation becomes residual too. This does not mean every local equation immediately fails at the horizon. It means that the licensing support for those equations is no longer same-scope standing-bearing in the old domain.

## Governance-depth

Metric depth is not primitive. It presupposes metric comparison. Once metric comparison is residual, depth can remain useful only while residual metric coherence remains high.

### Definition

Governance-depth is defined by residual successor-support thinning, not by metric depth. For role  $r$ , let

$$Succ_D^{res}(x, r)$$

be the class of post-break continuations of  $x$  that retain residual coherence of type  $r$  without restoring standing. Define

$$GDepth_D(x, r) \preceq GDepth_D(y, r) \quad \text{iff} \quad Succ_D^{res}(y, r) \subseteq Succ_D^{res}(x, r)$$

up to residual-coherence equivalence. Higher governance-depth means thinner residual successor support for the relevant role family.

**Lemma (Governance-Depth Is Not Metric Depth).** Governance-depth is not radial distance, proper time, affine parameter, curvature magnitude, or any metric scalar. It is a diagnostic of residual successor-support thinning:

$$GDepth_D(x, r) \text{ is defined by } Succ_D^{res}(x, r),$$

not by

$$r, \quad \tau, \quad \lambda, \quad R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}.$$

**Proof.** Radial distance, proper time, affine parameter, and curvature scalars are metric projections. Governance-depth is defined even when those metric descriptors are residual or failing, because it tracks the availability of role-preserving residual successor structure. Therefore it cannot be identical to a metric scalar. QED.

Governance-depth is schematic unless a particular carrier model supplies the relevant successor relation. The theorem requires the ordering of residual support loss, not a numerical construction of interior depth.

### Governance-depth and metric depth

Near the horizon, residual metric coherence may be high enough that metric depth approximates governance-depth. The ordinary phrase “deeper inside” remains useful. At intermediate governance-depth, metric depth becomes increasingly unstable: the coordinate or radial notion remains but

requires caution. At high governance-depth, the metric descriptor loses support and becomes skin or fails.

### Depth Failure Theorem

**Theorem (Metric Depth Failure).** Metric depth remains meaningful only while residual metric-comparison successor support remains nonempty. If residual metric-comparison successor support is exhausted, then metric depth is no longer a licensed residual descriptor:

$$Succ_D^{res}(x, metric) = \emptyset \Rightarrow \neg Depth_D^{res}(x).$$

**Proof.** Depth requires metric comparison. After the standing break, metric comparison is residual rather than fully licensed. Residual metric depth therefore depends on the existence of residual metric-comparison successor support. If  $Succ_D^{res}(x, metric)$  is empty, no continuation remains that preserves residual metric comparison. The descriptor “deeper” then no longer tracks a stable relation; it is, at most, skin or a formal coordinate artifact. This does not mean there is no place there. It means the descriptor has lost residual metric-comparison support. QED.

### Metric depth and governance-depth are distinct

**Theorem (Metric Depth and Governance-Depth Are Distinct).** AASC-GR may supply metric depth descriptors inside a licensed projection, but those descriptors remain standing-bearing only while the relevant metric-comparison and standing-continuation supports remain licensed. If residual metric successor support exhausts, governance-depth may invalidate the role of “deeper” as a descriptor even though local metric notation can still be written.

**Proof.** Metric depth is a descriptor inside the licensed metric projection. Standing-bearing use of that descriptor requires the metric-comparison role and the relevant continuation support. After the standing break, metric comparison is residual for crossed content. If residual metric successor support exhausts, the descriptor can remain as notation but no longer carries stable role-support. Therefore metric depth and governance-depth are distinct. QED.

At high governance-depth, metric descriptors can fail first as standing-bearing measurements before they fail as writable notation. A metric may remain syntactically expressible while its measurement role is already residual.

### Horizon, singularity, and depth

The horizon is the standing-break boundary for the uncertified case, not necessarily the place where residual coherence exhausts. A singularity, if represented, can be classified as an exhaustion signature of residual metric coherence. But this paper does not claim to compute a singular endpoint or replace it dynamically. It classifies the shift: classical depth toward a singularity becomes structural governance-depth toward residual-coherence exhaustion.

## Mergers and residual separability

A black-hole merger may produce a single exterior metric/radiative ledger and a post-merger exterior quotient. That exterior coalescence is not an AMetric collision. In the residual carrier, two formerly distinct horizon-isolated burdens may become jointly described by the post-merger exterior ledger, and their residual separability may thin or fail as a metric/comparison role. But when metric separability fails, the right classification is not that two AMetric regions touched. Contact, adjacency, distance, and collision are metric or causal roles. If the support for those roles is exhausted, the description loses the predicate required to state a collision.

Thus residual coalescence is an exterior or residual-ledger effect, not AMetric fusion. Endpoint tokens exposed by unresolved residues remain token-distinct no-crossing boundary statuses unless a separately declared shared governance domain licenses a common metric and interaction. Two realized interiors can collide only inside such a shared realized domain; AMetric boundary tokens merely coexist in the classification register.

## Part V - Measurement, Metrology, and Metric Standing

### Measurement is not primitive

Measurement is not a primitive event of contact between an object and a device. In AASC, a standing-bearing measurement is a governed role package. It requires comparison, scale anchoring, calibration continuity, target reference, witness or record fixation, quotient stability, and re-fixability. A local metric expression may remain writable inside a licensed AASC-GR projection, but writability is not measurement standing.

This distinction is required for the interior theorem. AASC-GR may preserve local metric notation, tensor components, source-ledger form, and local equations. None of those facts by itself licenses a standing-bearing measurement of crossed content inside an uncertified horizon-isolated carrier. Measurement inherits the standing status of the metrological envelope on which it depends.

The dependence on witness and record structure is not circular. A record can serve as a standing-bearing witness only when its own reference, calibration, role fixation, and re-identification conditions are already licensed in the declared scope. Measurement does not create record standing from nothing; it uses standing-capable record or witness structure as part of the governed metrological package.

### Standing-bearing measurement

**Definition (Standing-bearing measurement).** A standing-bearing measurement in a domain  $D$  is a role package

$$\text{Meas}_D(x, q)$$

where  $x$  is the target and  $q$  is the measured quantity, such that all of the following are licensed: metric or comparison carrier, scale anchor, calibration continuity, reference fixation for  $x$ , witness or record fixation for the result, quotient stability under permitted redescription, re-fixability or

admissible witness-equivalence in the declared governance scope, and standing preservation across the measurement act.

Thus

$$\begin{aligned} \text{Meas}_D(x, q) \Rightarrow & \text{MetricCompare}_D(q) \wedge \text{ScaleAnchor}_D(q) \wedge \text{Calib}_D(q) \\ & \wedge \text{Ref}_D(x) \wedge \text{WitnessRecord}_D(q) \wedge \text{QuotStable}_D(q). \end{aligned}$$

The implication is one-way. A metric expression or a displayed value can occur without satisfying the whole measurement package. A number inside a coordinate chart, a local tensor component, or a formal proper-time parameter is not automatically a standing-bearing measurement.

### Measurement residualization after metric standing loss

**Theorem (Measurement Residualization after Metric Standing Loss).** Let  $E_{\text{int}}$  be an uncertified horizon-isolated carrier for crossed content  $x$ . Suppose no certificate, leakage route, non-sink quotient, calibration bridge, witness ledger, or declared new governance scope is supplied. If metric comparison lacks full same-scope standing for  $x$  in  $E_{\text{int}}$ , then no measurement of  $x$  in the old domain  $D$  retains full same-scope standing. Where measurement-like organization remains representable, it has the status of residual metrological coherence rather than standing-bearing measurement.

Formally,

$$\neg \text{Stand}_D(\text{MetricCompare}_{E_{\text{int}}}(x)) \wedge \neg \text{Cert}_H \wedge \neg \text{NewGov}(D') \Rightarrow \neg \text{Stand}_D(\text{Meas}_{E_{\text{int}}}(x, q)).$$

Where representable,

$$\text{Meas}_{E_{\text{int}}}(x, q) \rightsquigarrow \text{ResCoh}_D^{\text{measurement}}(x, q).$$

**Proof.** A standing-bearing measurement requires a comparison carrier and the metrological supports that make the comparison re-fixable or admissibly witness-equivalent, calibrated, witnessed, and reference-fixed. In the uncertified horizon-isolated carrier, metric comparison for crossed content lacks full same-scope standing unless a certificate or new governance supplies the missing support. If the required comparison role lacks standing, the measurement role cannot retain full standing. A remaining measurement-like pattern may still be describable if it inherits pre-break metric, scale, calibration, or record organization. Such a pattern is not standing-bearing measurement; it is residual metrological coherence. QED.

### Metric writability is not measurement standing

**Corollary (Metric Writability Is Not Measurement Standing).** The fact that a local metric expression, scalar, tensor component, proper-time parameter, or source-ledger expression remains

writable in the AASC-GR projection does not by itself establish a standing-bearing measurement in the horizon-isolated carrier:

$$\text{Writable}_D(g_{\mu\nu}, T_{\mu\nu}, \tau, R) \not\Rightarrow \text{Stand}_D(\text{Meas}).$$

The corollary is not a claim that local notation is useless. It says that notation and measurement standing have different governance burdens.

## Measurement-tied role residualization

**Definition (Measurement-tied role).** A role is measurement-tied when its old same-scope standing depends on metric or comparison access, scale anchoring, calibration continuity, witness or record fixation, reference continuity, or re-fixability/admissible witness-equivalence. This definition is dependency-relative: it does not say that every role is measurement-tied in every scope, and it does not turn local apparatus dynamics into standing-bearing measurement by itself.

**Corollary (Measurement-Tied Role Residualization).** Any measurement-tied role becomes residual or governance-deficient for crossed content when measurement standing fails. This includes length, duration, proper-time assignment as a re-identifiable quantity, local energy measurement, mass/inertia measurement, temperature, entropy assignments tied to record status, curvature measurement, particle localization, field amplitude as a measured quantity, causal timing, and information extraction.

These roles are not declared physically absent. The claim is that their old same-scope measurement standing is absent unless a certificate, calibration bridge, witness ledger, or new governance supplies the missing metrological support.

## Calibration bridge requirement

**Corollary (Calibration Bridge Requirement).** If a measurement inside  $E_{\text{int}}$  is claimed to retain same-scope standing for exterior or crossed content, then at least one of the following must be supplied: a certificate, leakage or radiation witness channel, non-sink quotient, calibration bridge, boundary-to-boundary record map, declared new governance scope, or richer theory with explicit standing bridge. Without such support, the measurement claim is hidden governance import or bookkeeping.

## Residual metrological coherence

**Definition (Residual metrological coherence).**  $\text{ResCoh}_D^{\text{measurement}}(x, q)$  holds when  $x$  had pre-break measurement ancestry with respect to  $q$ , standing-bearing measurement fails after horizon isolation, some measurement-like organization remains representable, the retained organization is traceable to pre-break metric/scale/calibration/witness structure, and the result lacks full same-scope re-fixability or admissible witness-equivalence.

Residual metrological coherence is the measurement analogue of residual metric coherence. It can explain why local measurement-like descriptions may remain useful while refusing to treat them as standing-bearing measurements in the old scope.

## Part VI - Matter, Energy, and Source Roles

### Matter is not primitive

Matter, in the realized physical interior architecture, is not a primitive occupant of a container. It depends on a chain of roles:

$$\Phi_D \rightarrow \Pi_D \rightarrow G_D \rightarrow M_D \rightarrow \Omega_D \rightarrow T_D \rightarrow Cau_D.$$

Here  $\Phi_D$  is field-carrier mediation,  $\Pi_D$  is particle persistence class,  $G_D$  is gauge role,  $M_D$  is mass/inertia deformation response,  $\Omega_D$  is vacuum baseline,  $T_D$  is temporal retained consequence ordering, and  $Cau_D$  is causal witnessed dependence.

### Matter after standing break

If standing is absent and no new governance is declared, the roles required for ordinary matter cannot remain fully licensed. Field carriers have residual carrier-pattern status where representable. Particle classes have residual localization status where representable or identity patterns. Gauge roles become ambiguous unless boundary-fixed again. Mass/inertia response becomes residual deformation behavior. None of these is ordinary standing-bearing matter in the old scope.

### Matter role residualization

A formerly matter-bearing system may remain highly coherent near the horizon. It may retain field structure, bound structure, and source-like response as residual coherence. A strongly bound system may have greater residual persistence than a diffuse or weakly anchored description. But this is not preservation of matter standing. It is residual matter-role organization.

### Energy is not primitive

Energy is also not primitive. In the AASC physical stack, energy-like quantities function as transport, source, deformation, continuation, or stress ledgers inside a governed physical interior. A scalar energy density or flux is meaningful only relative to the roles that allow it to be compared, conserved, transported, or coupled to metric response.

### Energy after standing break

In a standingless carrier, energy configurations, where representable for crossed content, have residual source-load status. Stress-energy may remain formally writable in a metric projection. But the standing conditions under which the energy ledger is fully licensed are absent. Therefore conservation language becomes residual as well. The paper does not say energy conservation is simply violated. It says the role-conditions that license energy-conservation ledgers no longer have full same-scope support for crossed content.



## Matter-Energy Residualization Theorem

**Theorem (Matter-Energy Residualization).** In a standingless carrier without new governance, any retained matter-energy descriptions have residual source, field, particle, transport, and deformation-coherence status. This is a loss of role licensing, not a claim that matter is dynamically annihilated, locally destroyed, or empirically observed to disappear at the horizon. Within AASC-GR,  $T_{\mu\nu}^{\text{closed}}$  remains the metric-facing source ledger of the licensed projection wherever closed-source conditions hold; the residualization claim concerns the standing status of crossed content in the uncertified horizon-isolated carrier, not a local stress-energy discontinuity.

Formally,

$$Stand_D = 0 \wedge \neg RPI_D(E_{\text{int}}, x) \Rightarrow MatterEnergy_D(E_{\text{int}}, x) \rightarrow \text{ResCoh}_D^{\text{source/field/deformation}}(x).$$

**Proof.** Matter depends on field carriers, particle persistence, gauge role, mass/inertia response, local continuation, and comparison. Energy depends on source, transport, deformation, and metric-facing ledgers. These roles belong to the realized physical interior chain and require standing-bearing governance for full licensing. In the standingless same-scope carrier, that governance is absent for crossed content. Therefore matter-energy cannot remain fully licensed in the old sense. If matter-energy remains describable, it is residual source, field, transport, particle, and deformation coherence. QED.

## Mass and inertia

Mass and inertia are especially important because they may seem like scalar properties that survive by themselves. The mass/inertia role says otherwise. Mass is an operative deformation-response profile of particle persistence; inertia is its transport-state restriction. Without stable particle identity, transport, comparison, and deformation response, mass/inertia cannot remain fully role-stable.

**Corollary (Mass/Inertia Residualization).** In a standingless carrier, mass and inertia, where representable for crossed content, have residual deformation-response status unless new governance re-anchors particle identity, transport, comparison, and source response.

**Proof.** Mass/inertia require the particle persistence class, field-carrier mediation, gauge-compatible presentation, bounded transport, and quotient comparison. These are not fully licensed in the standingless carrier. Therefore any remaining mass/inertia behavior is residual deformation response, not fully standing-bearing mass role. QED.

## Matter-energy role chain in detail

The matter-energy conclusion is the boldest part of the paper, so the dependency chain must be explicit.

Field-carrier mediation  $\Phi_D$  supplies the distributed carrier role. Without it, a field value is a label rather than a persistence mediator. Particle persistence  $\Pi_D$  supplies localized identity classes. Without it, there are excitations or marks but no stable particle role. Gauge structure  $G_D$  fixes

presentation equivalence. Without it, presentation variants either proliferate falsely or support exhaustion real distinctions. Mass/inertia  $M_D$  supplies deformation response. Without it, mass is a scalar label rather than operative resistance. Vacuum  $\Omega_D$  supplies the non-null no-particle baseline. Without it, no-particle structure support exhaustions into nullity or substrate import. Time  $T_D$  supplies retained consequence ordering. Causality  $Ca_D$  supplies witnessed dependence.

Energy as a source or transport ledger sits across this chain. It is not a primitive substance that can float free of field, source, transport, comparison, and metric response. A stress-energy tensor is a metric-facing extraction of source roles. A flux is a transport ledger. A mass-energy relation is a relation between deformation response and source/transport roles. Each requires governance.

After the standing break, the chain does not become nothing by default. But each link becomes residual. Field carrier has residual carrier-pattern status where representable. Particle class has residual localization status where representable. Gauge becomes residual or ambiguous presentation. Mass/inertia has residual deformation-response status where representable. Vacuum has residual-baseline status where representable. Time and causality have residual order-and-dependency status where representable. The term “matter-energy role-license failure” therefore means support exhaustion of full role licensing, not instantaneous physical annihilation.

### Source ledger continuation versus crossed-content standing

The closed source ledger of AASC-GR and the standing status of crossed content are distinct. The condition

$$\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0$$

is a metric-projection condition on the closed source ledger. It does not by itself supply standing transport for a crossed reference class at a permanent horizon endpoint:

$$\nabla^\mu T_{\mu\nu}^{\text{closed}} = 0 \not\Rightarrow \text{StandTransport}_D(x, H).$$

Thus source-ledger continuation can remain licensed while crossed-content standing fails in the old scope. Residual source coherence is the post-break role status of crossed content when source-role organization remains describable but standing transport is not certified.

The same distinction applies to measurement of source content. The closed source ledger is not identical to a standing-bearing measurement of that ledger:

$$T_{\mu\nu}^{\text{closed}} \neq \text{Meas}_D(T_{\mu\nu}).$$

A standing-bearing source measurement requires metric comparison, calibration, witness/record fixation, and reference continuity in addition to source-ledger form.

## Conservation-language caveat

It is tempting to ask whether energy is conserved in the standingless interior. In the present framework the sharper question is whether the energy-conservation ledger remains licensed. A conservation law requires identified quantities, comparison across stages, continuation rules, and a domain over which exchange is closed or declared. These are governance structures. In the standingless carrier, absent new governance, those structures are residual. Thus conservation language may remain approximately useful where residual source and metric coherence are high, but it does not have full old-scope standing.

This avoids an overclaim. The paper does not say energy conservation fails dynamically. It says that the governance conditions under which energy conservation is a standing-bearing statement are absent unless reintroduced.

## Part VII - Time and Causality

### Time as residual order

Physical time in the AASC realized interior is not primitive flow. It is ordered retained consequence. Retained consequence requires standing. If standing is absent for crossed content in the old scope, time cannot remain fully licensed as ordinary physical time for that content.

### Time Residualization Theorem

**Theorem (Time Residualization).** In a standingless carrier without new governance, temporal order persists only as residual order while residual role coherence supports it.

**Proof.** Physical time is retained consequence ordering. Retained consequence requires standing-bearing fixation. In the standingless carrier, standing is absent for crossed content in the old scope. Therefore full temporal role licensing is absent. If before/after structure remains representable, it is residual order supported by residual metric, source, and record coherence. QED.

### No experiential claim

The theorem does not say that an observer experiences time stopping, slowing, fragmenting, or becoming chaotic. It classifies the role-status of time in the standingless carrier. A richer theory may supply a phenomenology, but this paper does not.

### Causality as residual dependency

Causality is witnessed dependence across admitted order. It requires witnessable relation, reference stability, and temporal role support. Without standing-bearing witness structure, causality cannot remain fully licensed in the old scope.

### Causality Residualization Theorem

**Theorem (Causality Residualization).** In a standingless carrier without new governance, causal relations, where representable for crossed content, have residual dependency-pattern status while

witness, metric, and temporal coherence remain.

**Proof.** Causality is witnessed dependence, not primitive influence. Witnessed dependence requires standing-bearing witnesses, reference, and temporal order. In the standingless carrier these supports are absent or residual. Therefore causality cannot remain fully licensed. Any causal structure that remains is residual dependency. QED.

### High governance-depth

At high governance-depth, before/after, causal dependence, local propagation, and prediction language all become unstable. This is not because a new exotic force is asserted. It is because the role supports that normally license temporal and causal language have thinned or support exhaustion.

## Part VIII - Certificates, New Governance, and Re-Coupling

### Why re-governance matters

The strong theorem is not a refusal of all interior physics. It is a refusal of undeclared interior physics as same-scope continuation after standing break. A new theory may provide a new governance structure. If it does, the residual-coherence result is no longer the full story. The question becomes how that governance is typed.

A certificate differs from a disconnected interior theory. A certificate connects crossed content back to the old standing class. A disconnected interior theory may describe a new domain but does not preserve the old standing. Hidden governance import silently assumes what must be proved. Repair introduces standing after loss while calling it preservation. The classification is central because it prevents residual coherence from being mistaken for a universal no-interior ontology.

### Re-coupling routes

A leakage or radiation route re-couples by carrying exterior witnesses. A holographic route re-couples by assigning boundary standing to what appears interior in the metric projection. An island route re-couples by changing the reconstruction domain and supplying an algebraic map. A fuzzball-like route may avoid the standingless carrier by replacing the classical interior with microstate-dependent exterior structure. A remnant route re-couples only if it exposes a remnant-to-witness map. A final-state route is admissible only if the final state is declared as an anchor rather than imposed as repair.

These routes are not rejected. They show exactly how the strong theorem can be avoided: provide governance. The paper's claim is that absent such governance, ordinary interior physical role-status cannot simply be assumed.

### Re-Governance Exhaustion Theorem

**Theorem (Re-Governance Exhaustion).** Any asserted standing structure for  $E_{\text{int}}$  after standing break is exactly one of: old-scope certificate, declared new governance, disconnected interior theory, hidden governance import, post-hoc repair, bookkeeping, or residual coherence.

**Proof.** If the structure preserves old standing, it must provide a map to the old standing class; that is a certificate. If it supplies its own kernel in a new domain, it is declared new governance. If it does not map to the old class, it is a disconnected theory relative to the old problem. If it is undeclared, it imports hidden governance. If it is added after loss while being counted as preservation, it is repair. If it changes no role content, it is bookkeeping. If it remains non-standing but structured, it is residual coherence. These cases exhaust the statuses available under the kernel. QED.

## Part IX - Interior Regime Stratification

### Three residual regimes

The paper can now state the interior picture in a controlled way. The regimes are not dynamical phases and not exact radial shells. They are structural regimes of residual-coherence support.

#### Near-horizon residual regime

In the near-horizon residual regime, standing has already been broken for the crossed class, but residual coherence is high. Metric and matter descriptions remain approximately ordinary. Tensor-like load persists. Bound systems may retain identity-like organization. Skin-level descriptions begin to lose support first, but the local physical description may remain serviceable. This is the role-status version of near-horizon local metric regularity without a local-dynamical discontinuity claim.

#### Intermediate governance-deficient regime

At intermediate governance-depth, residual successor support thins. Skin is largely unreliable. Metric comparison becomes more context-dependent. Particle and field identity lose support. Source and energy ledgers become less stable. Causal and temporal descriptions remain useful only as residual structures supported by the remaining role coherence.

#### Residual successor-support exhaustion regime

At high governance-depth, residual metric comparison weakens enough that depth itself loses stable meaning. Matter-energy distinctions become highly residual. Field, particle, source, time, and causal categories no longer function as ordinary physical roles. The carrier may remain non-null, but its description as metric-matter interior is no longer fully supported.

#### Stratification table

| Regime       | Standing                 | Residual coherence | Metricity                      | Matter-energy                      | Depth               |
|--------------|--------------------------|--------------------|--------------------------------|------------------------------------|---------------------|
| Near horizon | absent for crossed class | high               | approximate residual metricity | near-normal residual matter-energy | useful projection   |
| Intermediate | absent                   | thinning           | governance-deficient           | role-residualized                  | unstable projection |

| Regime | Standing | Residual<br>coherence | Metricity            | Matter-energy        | Depth                   |
|--------|----------|-----------------------|----------------------|----------------------|-------------------------|
| Deep   | absent   | low/collapsing        | strongly<br>residual | highly<br>degenerate | loses stable<br>meaning |

The key point is that standing is absent in all three regimes for the crossed class. What changes is residual coherence.

## Robustness and survival of residual coherence

The robustness of the original standing description affects how long non-degenerate residual coherence remains viable. Robustness is not standing persistence. It is a property of the pre-break role organization that determines how much residual structure can remain after standing is gone.

### Robustness variables

Important variables include tensor-load strength, boundary anchoring before crossing, internal coupling and cohesion, gauge-invariant structure, source/deformation-role stability, record redundancy, and resistance to redescription loss.

### Robustness theorem

**Theorem (Robustness of Residual Coherence).** Within a fixed standingless carrier, residual coherence persistence is monotone with respect to pre-break role robustness.

**Proof.** Residual coherence is inherited from pre-break role organization. Greater robustness means more invariant, tensor-like, or source-like organization was present before standing broke. Once standing is absent, that organization cannot supply standing, but it can supply residual persistence. Weaker pre-break organization supplies less residual structure. Therefore residual coherence persistence follows robustness. QED.

### Examples

A coordinate label has weak residual persistence. A foliation has weak residual persistence. A gauge presentation without boundary-fixed content is ambiguous quickly. A bound object has stronger residual coherence because its field, source, and deformation roles were more integrated. A stress-bearing field configuration may retain residual organization longer still. But no example retains old-scope standing unless certified.

## Part X - Information, Certificates, and Re-Governance

### Information is not residual coherence

Information is not primitive. It requires encoding, carrier, equivalence relation, and witness structure. Residual coherence is not information. It may support a residual pattern that looks information-like, but without standing-bearing encoding and re-fixability it is not standing-bearing information.

## Interior information

The phrase “information inside” is ambiguous. If it means standing-bearing information relevant to the old exterior reference class, a certificate is required. If it means residual patterning inside a standingless carrier, then it is residual coherence, not information in the standing-bearing sense. If it means a new interior theory with its own encoding and witness rules, that is new governance.

## Interior Information Theorem

**Theorem (Interior Information Boundary).** Interior information is not standing-bearing information unless a governance structure supplies encoding, carrier, equivalence, and re-fixability. In the standingless carrier without such governance, information descriptions are residual bookkeeping at most.

**Proof.** Information requires a standing-bearing carrier and encoding relation. The standingless carrier lacks same-scope standing for crossed content. No new governance is declared. Therefore no standing-bearing information structure is available. Any remaining information-like description is residual bookkeeping or hidden governance import. QED.

## Certificates and re-governance

A certificate is a governance structure that makes an otherwise unsupported standing claim admissible. It may be leakage, radiation correlation, boundary reconstruction, holographic anchor map, complementarity correspondence, remnant map, non-sink quotient, or another declared mechanism. If a certificate exists, the standingless-interior theorem does not apply in the same way, because the carrier is re-coupled or re-governed.

## Declared new governance

A new governance scope  $D'$  may support an interior physics. Such a scope is not ruled out. It may contain its own admissibility, standing, reference, witness, and continuation rules. But it is not the old same-scope physical interior unless it is linked by a certificate. The paper’s strong theorem concerns the absence of such declaration.

A new governance scope is present only when a new kernel tuple is declared:

$$K_{D'} = (Adm_{D'}, Stand_{D'}, Ref_{D'}, Irr_{D'}).$$

If  $K_{D'}$  is linked to the old domain  $D$  by a standing-preserving map, it acts as a certificate. If it is not linked, it may be a disconnected interior theory, but it is not preservation of the old standing class.

**Theorem (Retyped Horizon Classification).** If a horizon  $H_{BH}$  is treated as an admissibility-generating boundary, then exactly one of the following holds: it is no longer the same-scope derivative horizon but part of a new domain  $D'$ , it supplies a certificate into  $D$ , it belongs to a richer-scope theory with explicit bridge conditions, it imports hidden governance, or it is repair/bookkeeping.

**Proof.** A horizon treated as a source of governance must either introduce its own kernel tuple, map standing back into the old domain, or do no admissibility-bearing work. If it introduces  $K_{D'}$ , it is a new domain. If it maps to  $D$ , it is a certificate. If it is embedded in a larger physical theory, it is richer-scope. If it asserts standing without declaration, it is hidden governance import. If it is introduced after standing loss as preservation, it is repair. If it changes no standing or residual role content, it is bookkeeping. These alternatives exhaust the retyping routes. QED.

## Re-Governance Classification Theorem

**Theorem (Re-Governance Classification).** Any claim that  $E_{\text{int}}$  has standing after standing break is exactly one of: old-scope certificate, declared new governance scope, disconnected interior theory, hidden governance import, post-hoc repair, or bookkeeping.

**Proof.** If the claim maps crossed content back to the old standing class, it is a certificate. If it supplies a new domain with its own kernel roles, it is a declared new governance scope. If it does not map to the old standing class, it is a disconnected interior theory relative to the old problem. If it supplies standing without declaration, it is hidden governance import. If it restores standing after loss while calling it preservation, it is repair. If it changes no standing-bearing or residual role content, it is bookkeeping. These cases exhaust the ways standing can be asserted after break. QED.

## Standard proposals as re-governance candidates

Holography may re-govern the interior by treating relevant content as boundary-standing from the start. Islands may reassign reconstruction domains and provide an algebraic certificate. Fuzzball-like accounts may replace the classical standingless carrier with exteriorly distinguishable microstate structure. Remnants require a remnant-to-witness map. These proposals are not rejected here. They are routes for avoiding the standingless residual-coherence conclusion by supplying new governance or certificate structure.

## Part XI - Objections and Replies

### Objection 1. Does this contradict GR?

No. The paper presupposes AASC-GR. It distinguishes local metric projection from standing-preserving reconstruction. The Einstein equation may remain the licensed local metric equation while standing transport across an uncertified permanent horizon still fails.

### Objection 2. The equivalence principle says nothing special happens locally at the horizon.

The paper does not dispute local metric regularity or make a local infalling-experience claim. The theorem concerns same-scope standing transport and reconstruction status of the horizon-isolated carrier. Local metric validity is not standing transport.



**Objection 3. Stress-energy is covariantly conserved, so matter-energy continues.**

Covariant conservation of the closed source ledger is a metric-projection condition. It does not by itself supply a standing-transport certificate for a crossed reference class at a permanent horizon endpoint. Source-ledger continuation and crossed-content standing are distinct roles.

**Objection 4. An infalling observer can measure locally.**

The theorem does not deny local notation, local parameterization, or local device-model description inside AASC-GR. It denies that such descriptions automatically carry old same-scope standing for crossed content in the uncertified horizon-isolated carrier. Local measurement-like structure requires a standing-bearing metrological bridge if it is to count as same-scope measurement rather than residual metrological coherence.

**Objection 5. If  $T_{\mu\nu}$  is defined, matter-energy has been measured.**

No.  $T_{\mu\nu}^{\text{closed}}$  is a source ledger in the metric projection. A standing-bearing measurement of source content requires metric comparison, calibration, witness/record fixation, and reference continuity. Ledger standing, tensor writability, and measurement standing are different roles.

**Objection 6. Measurement residualization means empirical incoherence.**

No. It means loss of old same-scope role licensing. A new governance scope may supply a different metrological regime. Without that, measurement claims are residual or bookkeeping.

The measurement claim is also not a record/measurement bootstrap. A record counts as a measurement witness only when the record itself has licensed reference, calibration, and role-fixation status. A measurement act cannot manufacture the witness standing it requires.

**Objection 7. Higher-curvature or quantum-gravity terms fix this.**

They may, if declared. AASC classifies them as high-curvature or effective tensor load, leakage or certificate route, non-sink quotient, holographic or richer governance, or new scope. They are not hidden same-scope repair.

**Objection 8. Standingless means nothing exists inside.**

No. Standingless does not mean null. The carrier may retain residual coherence. The paper distinguishes non-null residual organization from standing-bearing physical interiority.

**Objection 9. If standing is absent, how can anything be described?**

Description can continue as residual projection. A metric, measurement, matter, energy, or causal description may remain useful where residual coherence supports it. The point is that any such retained description no longer has ordinary same-scope standing-bearing licensing unless new governance is declared.

**Objection 10. The interior could have its own governance.**

Yes, but then the governance must be declared. If it connects to the old crossed class, it is a certificate. If it does not, it is a disconnected interior theory. If it is not declared, it is hidden governance import. The strong theorem applies only absent certificate or new governance.

**Objection 11. Standing is licensed after the AMetric boundary, so why not after the horizon?**

Standing is licensed after the AMetric boundary because the AMetric boundary is an admissibility-generating boundary. It is prior to metricity, causal order, scale, and comparison. The black-hole horizon has the opposite dependency order. It is a metric-interior structure and already presupposes the governed comparison structure that would be required to license standing. Treating it as a new licensing boundary declares a new scope or imports hidden governance. It is not same-scope horizon crossing.

**Objection 12. Is the black-hole interior a new AMetric region?**

No. The residual-coherence interior is not pre-metric. It is post-metric and governance-deficient. Metric structures were already licensed in the exterior/classical representation and then lost full same-scope governance support after the standing break. A new AMetric-like boundary inside the black-hole regime may be posited only as a new theorem object  $D'$ , not as an automatic consequence of horizon crossing.

**Objection 13. You define the horizon as derivative by fiat.**

No. The horizon is derivative because its same-scope identity depends on metric and causal comparison. If it is retyped as fundamental, it is no longer the same-scope horizon of the original metric-interior projection; it is a new governance boundary, certificate, or richer theorem object.

**Objection 14. Residual coherence is just whatever is left.**

No. Residual coherence requires pre-break role ancestry, standing loss, no new governance or certificate, post-break non-null representation, retained organization, and failure of full role licensing. A remaining representation without role ancestry is not residual coherence of that role.

**Objection 15. Governance-depth is radial distance renamed.**

No. Governance-depth is defined by residual successor-support thinning. It can diverge from radial distance, proper time, affine parameter, and curvature magnitude. Metric depth is only a residual projection while residual metric comparison remains supported.

**Objection 16. This is just a singularity theory.**

No. Singularity is a possible diagnostic of late residual metric exhaustion. The main object is the whole standingless residual-coherence carrier. The near-horizon and intermediate regimes are part of the analysis.

### Objection 17. This predicts interior chaos.

No. The paper does not predict a dynamical chaos process. It classifies role-support loss under governance deficiency. Terms such as support-loss and support-exhaustion terminology mean licensing failure or residual-coherence support loss, not physical time evolution unless a separate dynamics is supplied.

### Objection 18. This is speculative.

It is a theoretical AASC extrapolation, but not unconstrained speculation. It composes kernel, horizon, metric, RPI, mass/inertia, information, measurement, and arrow results into a new paper-level theorem object. Its claims are conditional on the AASC premises.

## Part XII - Main Theorem and Conclusion

### Main theorem package

In the theorem below, successor-support notation remains structural unless a specific carrier model supplies the corresponding successor relation. The theorem requires dependency-ordered thinning of residual support, not a numerical construction of depth.

**Theorem (Standingless Interior and Residual Coherence).** Let  $D$  be a fixed AASC-governed scope. Let  $E_{\text{int}}$  be the horizon-isolated carrier reached after an uncertified permanent-horizon transition. Let  $x$  be crossed content whose pre-horizon standing has been broken. Suppose no leakage, standing-transport certificate, non-sink quotient, declared high-curvature/effective load, or declared new governance scope is supplied. Suppose further that the local AASC Einstein projection may remain valid but does not self-certify standing transport, and that metric, measurement, matter, energy, temporal, and causal roles are downstream of the realized physical interior architecture. Then:

1.  $\text{Stand}_D(x) = 0$ .
2.  $x$  is not transported into  $E_{\text{int}}$  as standing-bearing continuation.
3.  $E_{\text{int}}$  does not create replacement standing for  $x$  in the same scope.
4.  $E_{\text{int}}$  cannot instantiate a same-scope standing-bearing physical interior for  $x$ .
5. Any remaining describable structure is role-indexed residual coherence  $\text{ResCoh}_D^r(x)$ .
6. Residual coherence persists according to pre-break role load and robustness.
7. Skin has weakest residual persistence.
8. Tensor-like load, where representable, retains stronger residual organization but not standing-bearing tensor content.
9. For crossed content lacking standing transport, and where role-indexed organization remains representable, metricity, measurement-tied quantities, matter-energy, mass/inertia, time, and causality have residual/governance-deficient status rather than full same-scope role licensing.
10. Where measurement-like organization remains representable, it has residual metrological coherence rather than standing-bearing measurement.
11. Governance-depth replaces metric depth as the structural diagnostic, and at sufficient governance-depth metric depth loses stable meaning.

**Proof.** By the horizon break hypothesis, crossed standing is lost in  $D$ . By bivalence, no partial standing remains. By the kernel's no-creation/no-repair discipline, standing is not created inside  $E_{\text{int}}$  as replacement. By the AASC-GR compatibility theorem, local metric continuation is not a standing-transport certificate. By the absence of certificate, leakage, non-sink quotient, declared high-curvature/effective load, or new governance, no re-coupling or new domain supplies standing. Therefore  $E_{\text{int}}$  is not a same-scope standing-bearing physical interior for  $x$ . If no non-null description remains, the case is null or exhaustion. If non-null description remains, it cannot be standing-bearing role structure. It is residual coherence. Since residual coherence is inherited from pre-break role organization, its persistence follows role support and robustness where role-indexed organization remains representable. Skin carried no independent load, so it has weakest residual persistence. Tensor-like load carried role content, so where representable it retains stronger residual organization. Metricity, measurement, matter-energy, time, and causality are downstream of governance; for crossed content lacking standing transport, they have residual status only where role-indexed organization remains representable. Measurement additionally requires scale anchoring, calibration, witness/record fixation, reference continuity, and re-fixability or admissible witness-equivalence; when metric comparison lacks standing, measurement-tied roles cannot retain full same-scope standing without a metrological bridge. Depth depends on metric comparison; when residual metric comparison fails, depth fails. QED.

## Arc handoff summary

The interior theorem ends with residual coherence and governance-depth, not with New AMetric Boundary exposure. What is exported forward is a standingless, non-null residual carrier together with the diagnostic that endpoint exposure occurs only when residual metric-comparison successor support is empty. The next paper alone converts that exhaustion condition into endpoint-side AMetric normal-form status.

## Conclusion

**Series-position conclusion.** The result exported by this paper is residual-coherence status for the uncertified horizon-isolated carrier, together with the structural notion of governance-depth. It does not itself expose the New AMetric Boundary. The next paper may use this result only when residual metric successor support exhausts.

The standingless interior is not a hidden world of ordinary geometry and matter, nor is it simple nothingness. It is a residual-coherence regime: a horizon-isolated carrier in which standing is absent, governance support is broken, and the physical roles normally stabilized by standing have residual status where role-indexed organization remains representable. This conclusion is compatible with AASC-GR. Local metric dynamics may remain the licensed minimal first-response projection, but local metric validity is not a certificate of standing transport.

The result applies not only to metricity in isolation but to the metrological envelope built on metricity. In the standingless carrier, measurement-tied roles do not retain old same-scope standing merely because local notation remains writable. Where such roles remain describable, their status is residual metrological coherence.

The central consequence is not that standing fades. Standing does not fade. Standing is bivalent and, in the uncertified permanent-horizon case, broken. What varies afterward is residual coherence. Skin-level descriptions lose support first because they had little independent role load. Tensor-like structures, where representable, retain stronger residual organization because they carried role content before the break. For crossed content lacking standing transport, metricity, measurement-tied quantities, matter, energy, mass, time, and causality no longer retain full same-scope role licensing because they are downstream of the realized physical interior architecture. At sufficient governance-depth, the descriptor “deeper” itself loses stable metric meaning.

This paper therefore strengthens the black-hole reconstruction result from the interior side. The exterior theorem says that an uncertified permanent horizon cannot be a continuing same-scope standing-preserving reconstruction object. The present theorem says what follows if the kernel is pushed inward: absent certificate or new governance, the horizon-isolated carrier is not an ordinary standing-bearing physical interior. It is residual coherence after standing.

The paper therefore does not install residual coherence by naming what remains. It derives the residual-coherence classification from the failure of three alternatives: standing transport, same-scope horizon licensing, and declared new governance. A horizon cannot license the governance it presupposes; a residue without role ancestry is not residual coherence; and a new interior physics requires a new kernel tuple or certificate. The resulting standingless carrier is strong but bounded: it is not ordinary interior physics, not an AMetric origin, not nullity, and not a dynamical prediction of interior decay. It is the conditional AASC status left when governance is broken and no admissible re-governance is supplied.

## Appendix A - Formal dictionary

**Standing.** Bivalent governance status for licensed identity-preserving use in a fixed scope:  $Stand_D(x) \in \{0, 1\}$ .

**Standing break.** Loss of re-fixable standing in the declared witness and continuation structure.

**AMetric licensing boundary.** A pre-metric, admissibility-generating boundary at which a governed domain first becomes possible.

**Derivative metric boundary.** A boundary defined inside an already licensed metric/interior projection. It presupposes governance and cannot license standing in the same scope.

**Horizon-as-reset fallacy.** The mistaken inference that because standing is initially licensed downstream of the AMetric boundary, a black-hole horizon can repeat that licensing.

**Post-metric governance deficiency.** A condition in which metric roles were once licensed but have residual status, where representable, after their governance support is broken.

**Horizon-isolated carrier.** The sector  $E_{\text{int}}$  in  $X_H = E_{\text{ext}} \cup E_{\partial H} \cup E_{\text{int}}$  on which the old standing-transport relation is not defined for the crossed class in the uncertified case.

**Same-scope interior governance.** A governance tuple  $G_{\text{int}}^D$  that supplies admissibility, standing, reference, irreversibility, witnesses, and continuation for crossed content without changing scope or functioning as hidden certificate.

**Residual coherence.** Non-standing persistence of formerly governed role organization after standing has been broken.

**Role-indexed residual coherence.**  $\text{ResCoh}_D^r(x)$ , where  $r$  ranges over skin, record, gauge, tensor, metric, measurement, field, particle, source, time, and causal roles.

**Role load.** Pre-break structural work performed by a role.

**Residual persistence.** Structural persistence of residual coherence under governance-depth.

**Metric continuation.** Continuation of the licensed metric projection according to local curvature/source equations.

**Standing continuation.** Continuation of a standing-bearing reference class under admissible reconstruction, witness, certificate, or quotient-preserving transport.

**Standing-bearing measurement.** A role package requiring metric or comparison carrier, scale anchor, calibration continuity, reference fixation, witness/record fixation, quotient stability, and re-fixability or admissible witness-equivalence.

**Measurement-tied role.** A role whose old same-scope standing depends on metric or comparison access, scale anchoring, calibration, witness/record fixation, reference continuity, or re-fixability/admissible witness-equivalence.

**Residual metrological coherence.**  $\text{ResCoh}_D^{\text{measurement}}(x, q)$ , the non-standing persistence of measurement-like organization inherited from pre-break metrological roles after measurement standing fails.

**Calibration bridge.** Declared structure that preserves or re-establishes metrological continuity between a measurement-like interior organization and a standing-bearing witness or scale ledger.

**AASC-GR compatibility.** The standingless-interior theorem presupposes the AASC Einstein projection and denies only that local metric projection self-certifies horizon standing transport.

**Governance-depth.** Successor-support thinning:  $G\text{Depth}_D(x) \preceq G\text{Depth}_D(y)$  iff  $\text{Succ}_D(y) \subseteq \text{Succ}_D(x)$ , up to residual-coherence equivalence.

**Role-license failure.** Loss of full role licensing, not dynamical annihilation.

## Appendix B - Role-support matrix

| Role               | Pre-break basis              | Residual form               | Strong failure             |
|--------------------|------------------------------|-----------------------------|----------------------------|
| Coordinate skin    | chart over metric role       | label residue               | coordinate meaning fails   |
| Foliation skin     | slicing of metric/order role | residual slicing convention | simultaneity loses support |
| Gauge presentation | boundary-fixed equivalence   | ambiguous presentation      | gauge artifact             |

| Role               | Pre-break basis                                           | Residual form                   | Strong failure                                     |
|--------------------|-----------------------------------------------------------|---------------------------------|----------------------------------------------------|
| Record             | witness fixation                                          | residual bookkeeping            | no re-fixability or admissible witness-equivalence |
| Measurement        | metric comparison, scale, calibration, witness, reference | residual metrological coherence | no standing-bearing measurement                    |
| Information        | encoded standing-bearing carrier                          | residual pattern                | no standing-bearing information                    |
| Curvature response | metric tensor role                                        | residual curvature description  | invariant support loss                             |
| Field carrier      | persistence mediation                                     | residual field pattern          | carrier diffusion                                  |
| Particle class     | localized persistence                                     | residual localization           | identity-class failure                             |
| Mass/inertia       | deformation response                                      | residual response               | source ambiguity                                   |
| Energy/source      | transport/source ledger                                   | residual load                   | conservation-ledger failure                        |
| Time               | retained consequence                                      | residual order                  | before/after instability                           |
| Causality          | witnessed dependence                                      | residual dependency             | causal ambiguity                                   |

## Appendix C - Re-governance taxonomy

| Interior claim                              | Classification               |
|---------------------------------------------|------------------------------|
| Interior preserves old standing             | Certificate required         |
| Interior has new standing                   | New governance scope         |
| New governance maps to old class            | Certificate/re-coupling      |
| New governance does not map to old class    | Disconnected interior theory |
| Standing is assumed without declaration     | Hidden governance import     |
| Standing is assigned after loss             | Repair                       |
| Symbolic label is added with no role effect | Bookkeeping                  |
| Residual structure remains without standing | Residual coherence           |

## Appendix D - Source dependency ledger

| Source family    | Local use in this manuscript                         | Not imported      |
|------------------|------------------------------------------------------|-------------------|
| Kernel/bivalence | standing is bivalent; no standing creation or repair | physical dynamics |

| Source family               | Local use in this manuscript                                                                                                                                     | Not imported                                              |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| AMetric boundary            | initial standing licensing is boundary-level and pre-metric; the black-hole horizon is not such a boundary                                                       | claim that the black-hole horizon is a new AMetric origin |
| Horizon obstruction         | $E_{\text{int}}$ , partial transport, no uncertified horizon standing route                                                                                      | Hawking spectrum                                          |
| Realized physical interior  | metric, matter, energy, time, causality as downstream roles                                                                                                      | numerical parameter values                                |
| AASC Einstein projection    | licenses minimal local first-response metric dynamics; distinguishes local metric validity from standing transport                                               | all gravitational dynamics or endpoint recovery           |
| Measurement/record closure  | measurement standing requires metric comparison, scale, calibration, witness and record fixation, reference, and re-fixability or admissible witness-equivalence | local apparatus dynamics or collapse ontology             |
| Mass/inertia                | mass as deformation-response role                                                                                                                                | numerical mass spectra                                    |
| Arrow/time                  | time as retained consequence; causality as witnessed dependence                                                                                                  | thermodynamic mechanism                                   |
| Information non-primitivity | information requires carrier, encoding, witness                                                                                                                  | denial of information theory                              |
| Failure signatures          | singularity as diagnostic/exhaustion                                                                                                                             | singularity resolution                                    |

## Appendix E - Nonclaim ledger

This paper does not claim that the black-hole interior becomes a new AMetric boundary, that the horizon licenses standing in the same scope, that standing decays, that matter is annihilated at the horizon, that metricity instantly disappears, that nothing exists inside, that infalling observers experience a particular process, that GR is false, that the Einstein equation fails locally at the horizon, that local stress-energy conservation is standing transport, that quantum gravity is solved, that higher-curvature or quantum-gravity completions are denied when declared, that singularities are dynamically replaced, that residual coherence is a new substance, that governance-depth is a physical radial coordinate or curvature magnitude, that measuring devices physically explode or malfunction at horizon crossing, that local tensor notation becomes unwritable, that no observer can write a local parameter, or that measurement-like residual organization is impossible.



## Appendix F - Boundary comparison and no-reset normal form

| Feature                          | AMetric boundary                            | Black-hole horizon                                            |
|----------------------------------|---------------------------------------------|---------------------------------------------------------------|
| Dependency level                 | upstream of metricity                       | downstream of metricity                                       |
| Structural role                  | admissibility-generating boundary           | derivative metric boundary                                    |
| Relation to standing             | licenses first standing-bearing domain      | can obstruct or filter standing transport                     |
| Requires metric comparison       | no                                          | yes                                                           |
| Can generate same-scope standing | yes, as boundary-level governance admission | no                                                            |
| If treated as standing source    | ordinary AASC origin                        | new scope, certificate, hidden import, repair, or bookkeeping |

The no-reset normal form is:

$$B_{\text{AMetric}} \Rightarrow \text{governance admission}, \quad H_{\text{BH}} \not\Rightarrow \text{same-scope governance admission}.$$

If a black-hole interior is claimed to be newly standing-bearing, the claim must be typed as one of the following: declared new governance, certificate, richer-scope theory, hidden governance import, post-hoc repair, or bookkeeping. It is not a consequence of ordinary horizon crossing.

## Appendix G - AASC-GR compatibility ledger

| GR/AASC object                  | Role in Einstein projection                                               | Role in this paper                                                                               |
|---------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| $g_{\mu\nu}$                    | licensed metric comparison                                                | not a standing certificate by itself                                                             |
| $G_{\mu\nu}$                    | minimal first-response curvature response                                 | local metric equation remains compatible                                                         |
| $T_{\mu\nu}^{\text{closed}}$    | closed source ledger                                                      | source conservation does not imply crossed-content standing                                      |
| $\Lambda g_{\mu\nu}$<br>Horizon | homogeneous vacuum ledger endpoint/boundary role inside metric projection | not interior standing recovery requires certificate, leakage, non-sink quotient, or richer scope |
| Higher-curvature terms          | declared effective or richer tensor load                                  | possible declared route, not hidden repair                                                       |

The local metric projection and the horizon standing-transport problem have different targets. A metric equation can remain the licensed local description while crossed content lacks standing-preserving continuation. Conversely, a declared higher-curvature, holographic, island, or quantum-

gravity mechanism may avoid the standstill conclusion only by supplying a certificate, non-sink quotient, leakage route, or new governance scope.

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